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Carbon credit returns under EU ETS and its determinants: a multi-phase study

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Abstract

Purpose – The purpose of this paper is to find out the best method for forecasting European Union Allowance (EUA) returns and determine its price determinants. The previous studies in this area have focused on a particular subset of EUA data and do not take care of the multicollinearities. The authors take EUA data from all three phases and the continuous series, adopt the principal component analysis (PCA) to eliminate multicollinearities and fit seven different homoscedastic models for a comprehensive analysis.

Design/methodology/approach – PCA is adopted to extract independent factors. Seven different linear regression and auto regressive integrated moving average (ARIMA) models are employed for forecasting EUA returns and isolating their price determinants. The seven models are then compared and the one with minimum (root mean square error is adjudged as the best model.

Findings – The best model for forecasting the EUA returns of all three phases is dynamic linear regression with lagged predictors and that for forecasting EUA continuous series is ARIMA errors. The latent factors such as switch to gas (STG) and clean spread (capturing the effects of the clean dark spread, clean spark spread, switching price and natural gas price), National Allocation Plan announcements events, energy variables, German Stock Exchange index and extreme temperature events have been isolated as the price determinants of EUA returns.

Practical implications – The current study contributes to effective carbon management by providing a quantitative framework for analyzing cap-and-trade schemes.

Originality/value – This study differs from earlier studies mainly in three aspects. First, instead of focusing on a particular subset of EUA data, it comprehensively analyses the data of all the three phases of EUA along with the EUA continuous series. Second, it expressly adopts PCA to eliminate multicollinearities, thereby reducing the error variance. Finally, it evaluates both linear and non-linear homoscedastic models incorporating lags of predictor variables to isolate the price determinants of EUA.

Keywords ARIMA, Kyoto Protocol, Dynamic linear regression, European Union Allowance (EUA), European Union Emission Trading Scheme (EU ETS)

Paper type Research paper

1. Introduction

1.1 European Union Emissions Trading Scheme (EU ETS)

The EU ETS was envisaged after drawing lessons from the US Sulfur Dioxide Trading System (Alberola *et al.*, 2008), and consequently, incorporating decentralized decision making (Kruger *et al.*, 2007; Meyer, 2000) for allocation/auction of emissions allowances and proper monitoring of emitting sources. The European Union Allowance (EUA) gives an owner the right to release one ton of CO_2 in the atmosphere. The EU ETS covers carbon-intensive installations with a rated thermal input exceeding 20 MW across 30 countries (European Union's 27 member states, Norway, Liechtenstein and Iceland) and 41 percent of the Europe's CO_2 emissions. Each installation, out of total 12,000 installations covered under EU ETS, needs to surrender each year a number of allowances equal to its verified emissions and this number is fixed (Alberola *et al.*, 2009) by each member state in its National Allocation Plan (NAP). The EU ETS includes power and heat generation sector, oil refineries, factories making cement, glass, lime, bricks and ceramics, combustion plants, coke ovens, iron and steel plants, etc. The EU ETS is envisaged to proceed in three phases.

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The trial phase (Phase I) went from 2005 to 2007, and is like a "warm-up" period before the full-fledged implementation of the Kyoto Protocol. This has succeeded in sensitizing the companies regarding environmental issues by prising the carbon. Phase I suffered from the terrible problem of "over-allocation," leading to low carbon price and volatility due to excess supply of EUAs in the market. To counter this, inter-phase banking of allowances was introduced from 2008, when Phase II started, and allocation plans were tightened to deal with surplus supply and lack of demand issue.

Phase II (2008-2012) went alongside with the Kyoto Protocol and the third phase from 2013 onwards aims to reduce emissions by 20 percent (Chevallier, 2012) by 2020 (along with increasing energy efficiency by 20 percent and increasing the share of renewables in the energy mix to 20 percent, hence the "20-20-20" target). In total, 2.2 billion allowances per year during Phase I (2005-2007) and 2.08 billion allowances per year during Phase II (2008-2012) have been distributed. With a current value of around 8.5 euro per allowance, EU ETS (trial phase and phase II) has created a wealth of around 17 billion euros per year (Chevallier, 2012). The EU ETS has been expanded in Phase III to include PFC emissions from the aluminum sector; nitrous oxide emissions from acid production; and ammonium, ammonia and petrochemicals sectors. Aviation emissions were also included in 2012 for all flights flying in the European zone. The EU ETS has been marred by frauds like registry phishing and value-added tax frauds. However, good degree of liquidity achieved through financial intermediaries has so far prevented crimes like insider trading, etc. EU is contemplating to make the emission reduction targets more ambitious as most of the companies have already banked huge surplus of allowances into Phase III. The concerns are that raising the emission targets would adversely affect the competitiveness unless other countries become parties to the agreement and commit to comparable emission targets. Phase IV is being envisaged to commence on January 1, 2021 and finish on December 31, 2028. The European Commission, on January 22, 2014, proposed following two structural reform amendments:

(1) the linear reduction factor is to be increased from 1.74 percent (2013-2020) to 2.2 percent each year from 2021 to 2030, thus achieving the reduction target of 43 percent compared to the 2005 level; and

(2) a 12 percent "automatic set-aside" reserve mechanism of verified annual emissions is created in the fourth ETS period from 2021 to 2030, effectively as a carbon price floor with a price reset each year.

On October 24, 2014, the heads of EU member states approved the "market stability reserve (MSR)," which would change the amount of annually auctioned CO_2 allowances (Dhamija *et al.*, 2016) based on the amount of CO_2 allowances in the circulation. This would address imbalances in supply and demand by adjusting volumes for auction. The European Parliament and the European Council agreed on September 2015 for setting the starting date of the MSR to 2019. This mechanism puts the 900 million backloaded allowances in the reserve and reduces the reaction time of the MSR to one year.

1.2 EUA price development

BlueNext is the market dedicated to EUA allowances based in Paris. Likewise, European Climate Exchange (ECX) is the market based in London for future and options trading. NordPool, a common market for Finland, Norway, Sweden and Denmark is based in Oslo; the European Energy Exchange (EEX) based in Leipzig, trading spot and derivatives for emissions allowances and the New York Mercantile Exchange based in the USA, as well as trading futures and options are few other markets. The prices of these markets are strongly correlated, which is a standard feature of stock markets (Mansanet-Bataller and Pardo, 2009;

Galarraga *et al.*, 2013; Chevallier, 2012). The futures contract is a contract where each participant with an open position at the close of trading for that contract is obliged to make or take delivery of emission allowances to or from national registries. The trading unit is a lot of 1,000 emission allowances. Market participants may purchase consecutive contract months from December 2008 to 2020. Delivery takes place in the middle of the month before the contract expiration date.

Based on the demand and supply fundamentals, the factors identified by Christiansen *et al.* (2005), Mansanet-Bataller *et al.* (2007) and Alberola *et al.* (2008) as the price determinants for EUA EU ETS are: market fundamentals including the emissions-to-cap ratio, weather and production levels, the fuel switching and regulatory and policy issues. Mansanet-Bataller *et al.* (2007) and Alberola *et al.* (2008) analyzed the relations between energy markets and the EUA price. Based on Phase I data, Mansanet-Bataller *et al.* (2007) established that EUA prices are linked to the use of fossil fuel. Later, by using an extended data set, Alberola *et al.* (2008) further verify this relationship between energy and EUA prices. They find that this relationship varies depending on the period under consideration. In addition, Keppler and Mansanet-Bataller (2010) studied the causal relationship between EUA and electricity variables (such as clean dark spread, clean spark spread (CSS) and the switching price (SP)) during the first phase of the EU ETS. Benz and Trück (2009) used the GARCH models for the price dynamics for EUA. The price of EUA under EU ETS is strongly influenced by institutional decisions. Since, the participants fully react to the news arrival, any policy intervention from the regulator (EU Commission and specifically the director general for climate action) gets immediately reflected in the carbon price (Mansanet-Bataller and Pardo, 2009; Daskalakis and Markellos, 2009; Conrad *et al.*, 2012).

Chevallier (2012) analyzed the relationship between three energy variables (crude oil, coal and natural gas) and the EUA returns and found that all these energy variables are significant predictors of EUA returns. Power producers benefit by changing the proportion of coal and gas in their electricity mix depending on the prevailing price of the EUA. Switching enables them to reduce their emissions at lower cost. Depending on relative marginal abatement costs, it may be cheaper to switch to a less carbon-intensive fuel like gas instead of using coal and buying allowances. Following electricity variables are introduced in this regard.

Clean dark spread (CDS) (expressed in euro/MWh): It is the difference between the electricity price at peak hours ($elec_t$) and the coal ($coal_t$) price used to generate that electricity, corrected for the efficiency of the coal plant and the price of EUA (eua_t):

$$Clean\ dark\ spread_t = elec_t - \left(\frac{coal_t}{\rho_{coal}} + eua_t EF_{coal}\right) \tag{1}$$

where ρ_{coal} is the net thermal efficiency of a coal-fired plant (i.e. 35 percent according to Reuters), and EF_{coal} is the CO_2 emissions factor of a conventional coal-fired power plant (i.e. 0.95 tCO_2/MWh according to Reuters).

CSS (expressed in euro/MWh): it is the difference between the electricity price at peak hours ($elec_t$) and the natural gas (gas_t) price used to generate that electricity, corrected for the efficiency of the natural gas plant and the price of ERA (eua_t):

$$Clean\ spark\ spread_t = elec_t - \left(\frac{gas_t}{\rho_{gas}} + eua_t EF_{gas}\right) \tag{2}$$

where ρ_{gas} is the net thermal efficiency of a natural-gas-fired plant (i.e. 49 percent according to Reuters), and EF_{coal} is the CO_2 emissions factor of a conventional natural-gas-fired power plant (i.e. 0.41 tCO_2/MWh according to Reuters).

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SP (switch, expressed in euro/MWh): it represents the price at which both CDS and CSS become equal to:

$$Switching\ price_t = \frac{\frac{gas_t}{\rho_{gas}} - \frac{coal_t}{\rho_{coal}}}{EF_{coal} - EF_{gas}}. \tag{3}$$

As long as the EUA price is below the SP, coal plants are more profitable than gas plants and vice versa. Chevallier (2012) analyzed the relationship between CDS, CSS and SP and the EUA returns and found that CDS, CSS and SP are the significant predictors of EUA return series.

Extreme temperatures, constituting the most important dimension of weather (Roll, 1984), also impact the carbon price because they change the energy demand. For example, cold winters entail increased use of heating and hot summers entail extra need for air conditioning, and thus extra need for power generation. More power production entail more fuel use and hence more CO_2 emissions. This increases demand for CO_2 allowances, which in turn increases its price. So, a broad European temperature index can be treated as an appropriate carbon price driver.

All these studies analyze some subsets of the EUA data set and none of these existing studies takes into account the data from all the three phases of EU ETS. The current study expressly addresses this issue and extends the existing literature by comprehensively analyzing the data from all three phases of EU ETS and also the continuous series. The concatenated series (i.e. EUA continuous series) analysis, in addition to the separate phase analysis, can tell us about the stable price determinants. The current data from April 22, 2005 (launch date of EU ETS) till June 30, 2015 have been used in this study. This study comprehensively evaluates the price determinants falling into different categories, namely NAP events, energy prices like coal, oil and natural gas; extreme temperature events; and economic activity linkages such as stock exchange indices. In addition, none of the existing studies takes multicollinearities into consideration while analyzing the data. This may bias the standard errors thus obtained. The current study inspects the correlations between variables and adopts the principal component analysis (PCA) to avoid multicollinearities between variables. The current study comprehensively evaluates seven modeling techniques categories, namely linear regression (LR), dynamic linear regression (DLR), dynamic linear regression with lagged predictors (DLRLP), auto regressive integrated moving average (ARIMA), ARIMA errors (AE), ARIMA errors with lagged predictors (AELP) and ARIMA with transfer function method (ARIMAX) for forecasting EUA and assessment of significant price determinants for all the four data series, i.e. EUA Phase I, II, III and the continuous series. The seven models are then compared and the one with the minimum root mean square error (RMSE) is adjudged as the best model. Even within each individual modeling technique category, the study uses an iterative method to find out the best possible predictors and/or lags of predictors using the standard measures such as adjusted R^2 and Bayesian information criterion (BIC).

2. Objective

The objective of this study is to find out the best method for forecasting EUA and determine the relevant price determinants of EUA for all three phases of EU ETS and the continuous series.

3. Data

Choosing relevant data is not trivial, especially in a young market. The fundamentals' theory reveals the basic market mechanism, and the assets of interest are as follows: EUA price, electricity price, coal price, natural gas price and oil price. These are essentially

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all commodities. Along with these NAP announcement data, the data regarding extreme temperature (both minimum and maximum) and stock index data as a measure of economic activity are also relevant in this context.

3.1 Carbon credit data

The trading of EUA futures contracts is primarily performed through the ECX in the Netherlands. ECX uses Powernext spot prices as a reference for its futures contracts. From March 22, 2005 to the end of the sample study, i.e. June 30, 2015, we use the ECX December 2007, December 2012 and 2015 expiry contracts for phase analysis and the continuous (rolled back to 2005, i.e. price of December 2007 and 2012 expiry futures are adjusted to correspond with the series of December 2015 expiry future to form a concatenated data series) December 2015 contract for a continuous analysis. ECX futures price as shown in Figure 1 appears volatile. For the current analysis, three futures contracts that expire in the month of December of each phase end, i.e. 2007, 2012 and 2015, are used. The data correspond to the daily average of all intra-day quotes updates within a day. There are 682, 1,975, 1,263 and 2,625 observations for Phase I, Phase II, Phase III and continuous series, respectively. Finally, given that carbon prices are not stationary, they are converted into stationary returns by taking first logarithm differences. That is, we use continuously compounded returns constructed as $r_{c,t} = \ln(P_{c,t}/P_{c,t-1})$, where $P_{c,t}$ is the carbon price at time t. The financial economics literature provides evidence that most asset prices are not stationary and most asset returns are indeed stationary. Using the Dickey-Fuller method to test for stationarity of carbon returns and prices during our sample period, we find that carbon price series is not stationary, but returns indeed are. Therefore, for current analysis also, we use the same return data series. Although the Jarque-Bera test indicates that carbon returns are not normally distributed, its impact on the statistical inference is limited for the following reasons: our sample size is large and the test statistics used here have properties that ensure asymptotic normality. Box-Pierce test for EUA returns reveals that the returns of EUA Phase I, II and continuous series are autocorrelated, whereas those of EUA Phase III are not.

3.2 The announcement data

The announcement data are gathered from a combination of the European Commission official website, the Community Independent Transaction Log website and the Point Carbon news archives. From these three sources, the earliest date that each announcement is made public is taken. This is a marked improvement over the existing literature on the same issue and an attempt is made to account for the various NAPs and verifications information leakages prior to the official release dates throughout Phases I and II. There are 6, 27 and 26 (total 59) announcements for Phases I, II and III, respectively. Three dummy variables are created which mark the presence or absence of particular NAP phase events on a particular date.

3.3 The energy variables data

Most power companies purchase and hedge on a year-ahead basis. Therefore, in this study, front-year futures contracts are used. The data have been procured from Energy Market Price, Market Intelligence Group SA, Rue Du Rhone 5, 1921 Martigny, Switzerland (MIG, 2015). CIF API2 Index data are taken as coal data. The price of coal has to be first converted from $/ton to $/kg; then by adopting the coal heat of combustion (7,400 kcal/kg) it is possible to convert from $/kg to $/kcal and then to $/Gcal. 1 Gcal corresponds to 1.16 MWH and, in this way, the coal data are converted to €/MWH using the appropriate €/$ conversion rate. National Balancing Point (NBP) data are taken as the natural gas data. The price of gas is

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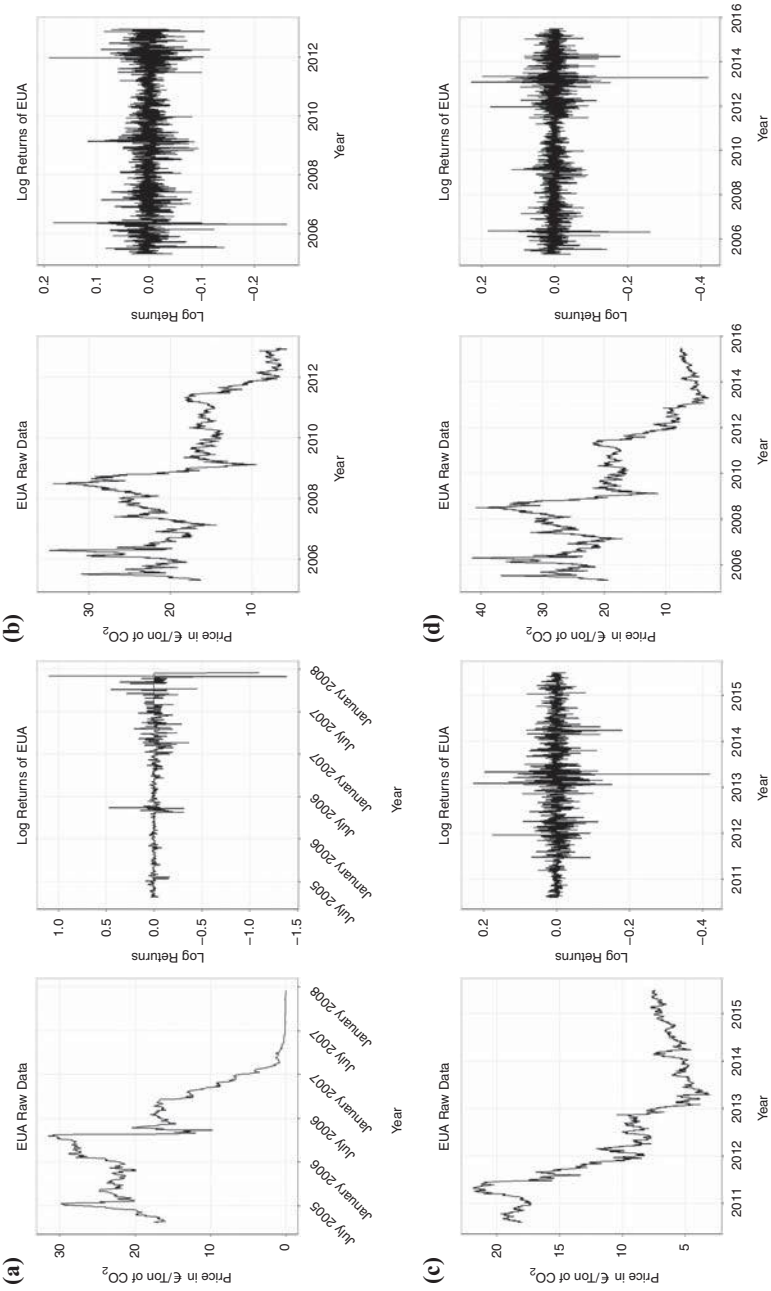

Notes: (a) EUA Phase I (April 22, 2005-December 17, 2007); (b) EUA Phase II (April 22, 2005-December 17, 2012); (c) EUA Phase III (August 6, 2010-June 30, 2015); (d) EUA Continuous (April 22, 2005-June 30, 2015)

Figure 1.
Raw data (left panels) and log returns (right panels) of EUA

first converted from GBpence/Therm to GBpence/British Thermal Unit (BTU). Using the relation 1 million BTU = 0.29 MWH, the price of gas is then converted to GBpence/MWH and then to £/MWH and finally to €/MWH using the appropriate €/£ conversion rate. Brent Crude oil data are taken as oil data. This is converted from $/BBL to €/BBL using the appropriate €/$ conversion rate. Finally, this is converted to €/MWH using a conversion factor of 1.6282. EEX peak load power (€/MWH) is used for electricity data as it is perceived that fuel switching occurs on the marginal unit of power produced, and that fuel mix for base power is relatively constant. In this way, all the energy data are converted to €/MWH. Finally, log returns series are calculated and used instead of the price data series themselves.

3.4 Extreme temperature data

Minimum and maximum temperature data are taken from European Climate Assessment & Dataset (Tank, 2015). Temperature data are averaged for 18 European countries, weighted by the weight of each country in the total volume of distributed allowances:

$$T = \frac{\sum_{j=1}^{18} Q_j \theta_j}{\sum_{j=1}^{18} Q_j} \tag{4}$$

where T is the monthly index, Q_j is the number of allowances allocated by country j in its NAP and θ_j is the national temperatures index of country j.

3.5 Economic indicators data

The DAX index data are taken from Quandl, 9 Prince Arthur Ave, Toronto, Canada (DTG, 2015) and the ticker "ECST" on Bloomberg. This is used as an indicator for the economic activity.

4. Model variables

4.1 Carbon price

In Figure 1(d), the carbon price shown is the continuous December 2015 futures contract for EUAs. It shows data from April 22, 2005 to June 30, 2015 with roll back occurred on December 31, 2007 and Decemeber 31, 2012. Figure 1(a) shows data for the EUA Phase I, i.e. December 2007 futures contract. It shows data from April 22, 2005 to December 17, 2007. Figure 1(b) shows data for EUA Phase II, i.e. December 2012 futures contract. It shows data from April 22, 2005 to December 17, 2012 with roll back occurred on December 31, 2007. Figure 1(c) shows data for EUA Phase III, i.e. December 2015 futures contract. It shows data from August 6, 2010 to June 30, 2015 with roll back occurred on December 31, 2012. In concatenating time series data for subsequent year-ahead futures, the connection point represents a discontinuity, and the corresponding return data point is removed.

Starting from 8 €/ton of CO_2 on January 1, 2005, EUA prices have risen to around 30 €/ton until the release of 2005 verified emissions. On April 24, 2006, EUA prices experienced a sharp break due to the first compliance disclosures by the member states (Alberola *et al.*, 2008). Verified emissions were about 80 million tons (or 4 percent) lower than the amount of allowances allotted to installations during the 2005 compliance period (Ellerman and Buchner, 2008). As the 2005 compliance confirmed that the allowance market is over-supplied and the European Commission reaffirmed on October 2006 its will to enforce tighter targets during Phase II, the EUA spot price and December 2007 futures price had been asymptotically declining toward zero, whereas the December 2008 futures price exhibited an increasing price pattern of 25 €/ton at the end of 2007. The disconnection between EUA prices of Phase I (2005-2007) and the prices of Phase II (2008-2012) is largely explained by the non-transferability of unused EUA Phase I allowance to the next phase,

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and this became evident after the 2007 compliance event that occurred on April 2008 (Alberola and Chevallier, 2009). The price development exhibited a lower bound below 10 €/ton of CO_2 in February 2009 (when financial crisis was going on), and an upper bound around 30 €/ton of CO_2 in May-June 2008 (just after to the 2007 compliance event). The total volume of transactions on the European carbon market (including exchange-based as well as over-the-counter trading) had grown from 262 million ton in 2005 to 5,016 million ton in 2009, and then declined to 3,313 million ton in 2015. As evident from the figures, all the EUA price series are non-stationary and all the EUA return series are stationary. We choose EUA return series to work with.

4.1.1 Descriptive statistics. In Table I, we can see that the mean carbon price over the period is equal to 12.75, 17.85, 9.74 euro/ton of CO_2 in Phases I, II and III, respectively and 17.44 euro/ton of CO_2 for continuous series EUA. These descriptive statistics of EUA show positive skewness and lower kurtosis, whereas EUA returns show negative skewness and excess kurtosis (for a normally distributed random variable skewness is 0, kurtosis is 3 and excess kurtosis is 0). The Jarque-Bera Test shows departure from normality for both EUA and EUA returns.

4.2 Institutional decisions

In order to evaluate the impact of news about Phases I, II and III NAPs by the European Commission on the price of carbon, total 59 announcements considered for this study are summarized in Table II.

The dummy variables refer to news information disclosure concerning NAPs Phase I (NAPI), Phase II (NAPII) and Phase III (NAPIII). The main sources for the news events recorded are UNFCCC, the European Commission, the European Parliament, the European Economic and Social Committee, ECX, EEX, Bluenext, ICE and Point Carbon. The period under consideration goes from March 22, 2005 to June 30, 2015, i.e. it covers a sample of 2,625 daily observations. During the days when no news event fell in either of the three categories, all the three dummy variables take the value of "0," otherwise dummy variable for the category in which the news event fell was coded as "1" and other two dummy variables were coded as "0."

4.3 Energy variables

The Brent price (expressed in Euro/BBL) is the Brent crude oil futures month-ahead price traded on ICE (Figure 2(a)). We can see that after declining in the year 2014, the prices have again started rising in 2015. Brent is a North Sea deposit and is representative of the crudes produced in this region. Therefore, it is the appropriate measure to match other energy variables traded in the continental Europe, which are also the determinants of the carbon price.

Table I. Descriptive statistics of EUA and EUA returns

	EUA Phase I	EUA Phase II	EUA Phase III	EUA Continuous	EUA return Phase I	EUA return Phase II	EUA return Phase III	EUA return Continuous
Mean	12.75	17.85	9.74	17.44	−0.01	−0.00	−0.00	−0.00
SD	10.38	6.28	5.24	9.27	0.11	0.03	0.03	0.03
Skewness	0.01	0.08	0.97	0.19	−3.12	−0.52	−1.32	−1.19
Excess kurtosis	−1.50	−0.53	−0.48	−1.03	74.43	8.45	22.35	20.01
Jarque-Bera	63.52	24.85	209.18	130.86	159,244.2	5,977.71	26,724.56	44,453.35
Probability	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Observations	682	1,975	1,263	2,625	682	1,975	1,263	2,625

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S. No.	Date	NAP announcements	Dummy variables NAPI	NAPII	NAPIII
1	May 25, 2005	Italy's Phase I allocations	1	0	0
2	June 20, 2005	Greece's Phase I allocations	1	0	0
3	September 15, 2005	Established EU ETS Scheme	1	0	0
4	February 22, 2006	Amendment for allocation of allowances	1	0	0
5	April 12, 2006	Amendment to allocation of allowances	1	0	0
6	October 11, 2006	UK – Temporary exclusion of installations	1	0	0
7	November 29, 2006	Phase II allocations for Germany, Greece, Sweden, Ireland, etc.	0	1	0
8	January 16, 2007	Belgium, Netherlands Phase II allocations	0	1	0
9	February 5, 2007	Slovenia Phase II allocations	0	1	0
10	February 26, 2007	Spain Phase II allocations	0	1	0
11	March 26, 2007	Czech and France Phase II allocations	0	1	0
12	April 2, 2007	Austria Phase II allocations	0	1	0
13	April 16, 2007	Hungry Phase II allocations	0	1	0
14	May 15, 2007	Italy Phase II allocations	0	1	0
15	June 4, 2007	Finland Phase II allocations	0	1	0
16	June 13, 2007	Sweden allocations	0	1	0
17	July 13, 2007	Ireland, Sweden, Luxemburg	0	1	0
18	July 18, 2007	Cyprus Phase II allocations	0	1	0
19	August 31, 2007	Denmark Phase II allocations	0	1	0
20	October 18, 2007	Portugal Phase II allocations	0	1	0
21	October 26, 2007	Germany, Romania Phase II allocations	0	1	0
22	December 7, 2007	Slovakia Phase II allocations	0	1	0
23	December 19, 2007	Liechenstein – EFTA surveillance authority decision	0	1	0
24	December 17, 2008	The Netherlands – Inclusion of additional gases	0	1	0
25	February 27, 2009	EFTA surveillance authority decision	0	1	0
26	September 7, 2009	Austria – Inclusion of additional gas	0	1	0
27	December 11, 2009	Poland, Estonia Phase II allocations – Rejected	0	1	0
28	April 19, 2010	Poland Phase II allocations	0	1	0
29	May 19, 2010	Emission of EU falls more than 11% in 2009	0	0	1
30	May 26, 2010	Move to 30% emissions cut	0	0	1
31	September 7, 2010	Aviation included in the scheme	0	0	1
32	October 22, 2010	Cap on 2013 emission cut adopted	0	0	1
33	November 5, 2010	Linking ETS with Sweden	0	0	1
34	December 17, 2010	Cancun Agreement	0	0	1
35	January 19, 2011	EU ETS registry system	0	0	1
36	February 23, 2011	Steps to combat registry security thefts	0	0	1
37	March 24, 2011	Estonia	0	1	0
38	April 29, 2011	Germany Phase II allocations	0	1	0
39	June 6, 2011	UK – inclusion of additional gas	0	1	0
40	November 16, 2011	ETS Registry New Rules	0	0	1
41	December 5, 2011	Estonia Phase II allocations	0	1	0
42	December 12, 2011	Durban Conference	0	0	1
43	February 3, 2012	Italy – inclusion of additional gases	0	1	0
44	March 26, 2012	New rules for verification of units	0	0	1
45	May 23, 2012	Temporary free allowance for Cyprus. Estonia, Lithuania	0	0	1
46	June 20, 2012	Single ETS registry activated	0	0	1
47	July 6, 2012	Temporary free allowance for Czech, etc.	0	0	1
48	July 23, 2012	Phase III auctions	0	0	1
49	December 10, 2012	Doha Conference	0	0	1
50	January 1, 2013	Free allocation of allowances	0	0	1
51	April 2, 2013	Verified emissions for 2008-2012	0	0	1

(*continued*)

Table II. National Allocation Plan announcements

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Table II.

S. No.	Date	NAP announcements	Dummy variables NAPI	NAPII	NAPIII
52	May 16, 2013	CERs and ERUs surrendered for Phase II	0	0	1
53	November 18, 2013	2014 auction calendar published	0	0	1
54	February 27, 2014	Back loading: 2014 auctions reduced by 400M allowances	0	0	1
55	April 1, 2014	Verified emissions for year 2013	0	0	1
56	April 30, 2014	Credits exchanged by April 30, 2014 announced	0	0	1
57	May 15, 2014	Compliance data for year 2013	0	0	1
58	April 1, 2015	Verified emissions for year 2014	0	0	1
59	May 4, 2015	Compliance data for year 2014	0	0	1

The natural gas price used (expressed in £/Therm) is the futures month-ahead natural gas price negotiated on NBP Hub (Figure 2(b)). It is the major liquid gas trading market in Europe along with the others like Henry Hub, Zeebrugge, etc., and has a major impact on consumer price of the gas in Europe. The coal price is the Antwerp/Rotterdam/Amsterdam (ARA) coal futures month-ahead price (Figure 2(c)), which is the major imported coal in northwest Europe. The ARA coal is expressed in euro/ton. The coal prices have been consistently declining after 2011, leaving the firms to purchase carbon credits rather than reducing carbon footprints. The CDS, CSS and the SP are shown in Figure 2(d)-(f), respectively. Whether the firms will switch to cleaner fuels like gas will ultimately depend on CDS and CSS. The log returns of all energy variables characteristically show the stationarity, whereas the raw energy variables do not. Therefore, we use log returns for our modeling purposes.

4.4 Extreme weather events

Figure 3 shows the maximum and minimum temperatures of the Euro zone as defined in Equation (4). For calculating the extreme maximum temperatures, the average ($\mu_{\max}$) and standard deviation ($\sigma_{\max}$) of the temperatures of the summer months, i.e. June-August for all the years are calculated and then extreme maximum temperature is defined as temperature $> \mu_{\max} + 1.5\sigma_{\max}$. The dashed horizontal lines in the graph stand for $\mu_{\max} - 1.5\sigma_{\max}$, $\mu_{\max}$ and $\mu_{\max} + 1.5\sigma_{\max}$. Similarly for calculating the extreme minimum temperatures, the average ($\mu_{\min}$) and standard deviation ($\sigma_{\min}$) of the temperatures of the winter months, i.e. December-February for all the years are calculated and then extreme minimum temperature is defined as temperature $< \mu_{min} - 1.5\sigma_{\min}$. The dashed horizontal lines in the graph stand for $\mu_{\min} - 1.5\sigma_{\min}$, $\mu_{\min}$ and $\mu_{\min} + 1.5\sigma_{\min}$. The impact of temperatures on carbon prices is essentially non-linear, i.e. it is possible to identify statistically significant effect of temperature events only above or below these threshold extreme temperatures. Two dummy variables for the extreme hot days and extreme cold days are created. The dummy variable for extreme hot temperature is assigned "1" for all days when temperature exceeded this mean +1.5 SD, otherwise value "0" is assigned and the dummy variable for extreme cold temperature is assigned "1" for all days when temperature was less than this mean −1.5 SD, otherwise value "0" is assigned.

4.5 Correlational analysis

In order to select the appropriate model variables, it is important to perform the correlational analysis because any correlation between variables which is more than 0.6 will cause multicollinearities, thus inflating the standard errors.

As evident from Tables III-VI, the data of all three phases and the continuous series show a high degree of correlation between SP returns and gas returns (GAS) and between CDS

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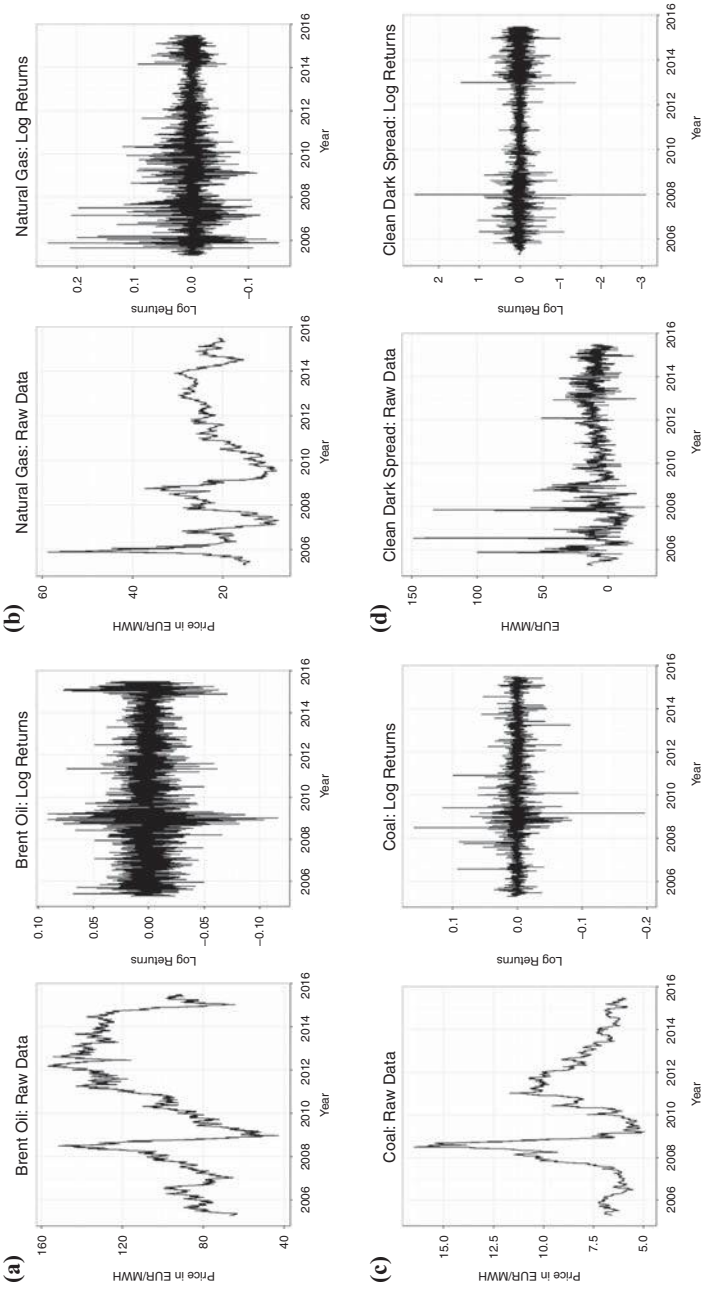

Figure 2. Raw data (left panel) and log returns (right panel) from April 22, 2005 to June 30, 2015 (continuous series from Phase II to Phase III)

(continued)

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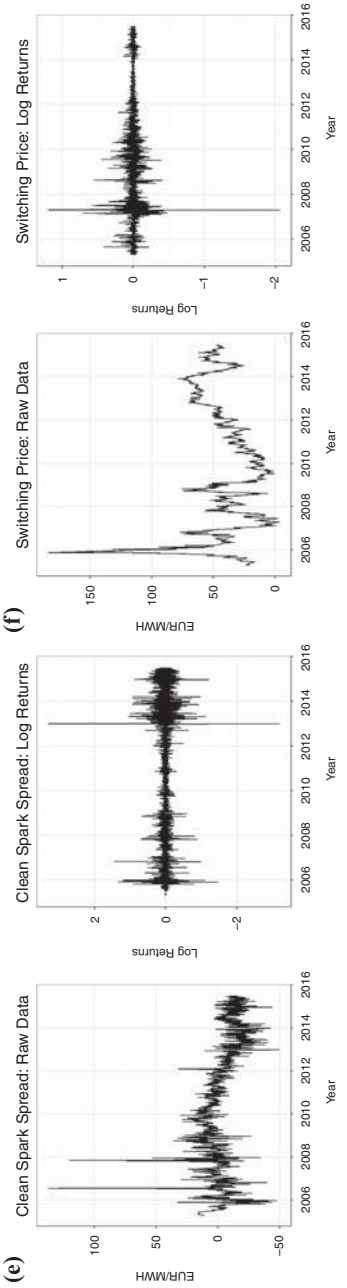

Note: (a) Brent; (b) natural gas; (c) coal; (d) clean dark spread; (e) clean spark spread; (f) switching price

Figure 2.

return and CSS return. Consequently, PCA is employed to get rid of multicollinearities so that regression can be done. Table VII for rolled over series data shows that variables CDS and CSS load high on factor "RC1," and GAS and SP load high on factor "RC2." The rest of the variables have one to one correspondence with the other factors. Similarly Tables VIII-X for the data of Phases I, II and III, respectively, show that variables CDS and CSS load high on factor "RC2" and variables GAS and SP load high on factor "RC1." It is worth noting that "RC1" and "RC2" (rotated components) are just symbolic names given to the identified factors by the "R" software. The RC1 is named as switch to gas (STG) and RC2 is named as clean spread (CS) for EUA Phases I, II and III. For the EUA composite series, RC2 is named as STG and RC1 is named as CS. The rest of the variables have one to one correspondence with the other factors. So for further regression analysis, we can use factors RC1, RC2 and the variables NAPI, NAPII, NAPIII, OIL, COAL, DAX, COLD, HOT. We use the first difference series of all these variables (since first difference of all these variables are found to be stationary) and named these series as RC1, RC2, napI, napII, napIII, brent, coal, dax, tempcold, temphot. So these ten data series are used to model all four differenced EUA series (represented at EUA), i.e. Phases I, II, III and continuous series.

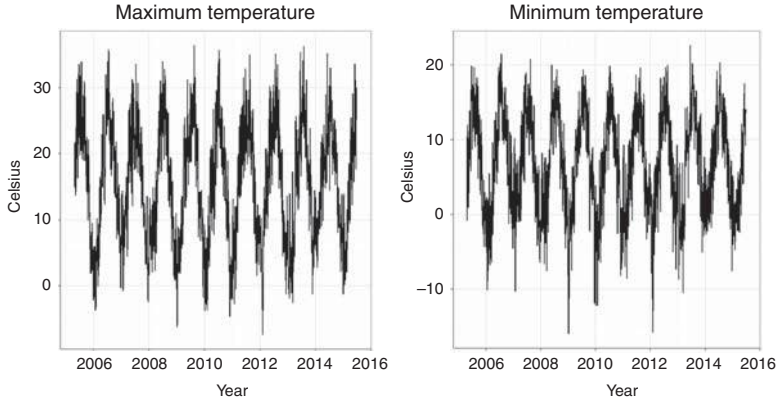

Figure 3. Maximum temperature (left panel and minimum temperature (right panel) of Euro Zone from April 22, 2005 to June 30, 2015

	EUA	NAPI	NAPII	OIL	GAS	COAL	CDS	CSS	SS	DAX	COLD
EUA											
NAPI	0.03										
NAPII	−0.05	−0.01									
OIL	0.12**	−0.02	0.06								
GAS	0.09*	−0.03	0.08*	0.15***							
COAL	0.00	0.01	0.05	0.02	−0.02						
CDS	0.00	−0.02	0.07	0.04	0.05	0.02					
CSS	0.00	0.00	0.04	−0.01	−0.16***	0.01	*0.79****				
SP	0.05	−0.01	0.06	0.06	*0.66****	−0.16***	0.00	−0.09*			
DAX	−0.02	−0.01	0.02	0.08*	0.04	−0.07	0.05	0.01	0.03		
COLD	−0.04	−0.03	−0.08*	0.02	−0.01	0.00	−0.03	−0.05	0.00	−0.03	
HOT	0.02	0.07	−0.03	−0.01	0.01	−0.02	0.00	0.00	0.01	−0.03	−0.12**

Notes: CDS, clean dark spread; CSS, clean spark spread; SP, switching price; HOT, COLD, extreme temperatures days. *,**,***Significant at 0.05, 0.01 and 0.001 levels, respectively

Table III. Correlation table – Phase I

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Table IV. Correlation table – Phase II

	EUA	NAPI	NAPII	NAPIII	OIL	GAS	COAL	CDS	CSS	SS	DAX	COLD
EUA												
NAPI	0.04											
NAPII	0.00	−0.01										
NAPIII	−0.02	0.00	−0.01									
OIL	0.17***	−0.01	0.02	0.01								
GAS	0.17***	−0.02	0.03	−0.01	0.11***							
COAL	−0.08***	0.01	0.00	0.01	0.02	0.00						
CDS	−0.01	−0.01	0.05*	0.00	−0.01	0.04	−0.03					
CSS	0.00	0.00	0.04	0.00	−0.03	−0.15***	0.00	*0.80****				
SP	0.13***	−0.01	0.02	−0.01	0.05*	*0.74****	−0.23***	0.02	−0.10***			
DAX	0.19***	0.00	0.02	−0.06*	0.28***	0.06**	−0.17***	0.03	0.01	0.07**		
COLD	−0.03	−0.02	−0.03	0.01	0.04	−0.01	0.03	−0.03	−0.04	−0.02	0.02	
HOT	−0.03	0.06*	−0.02	−0.01	0.00	0.00	−0.02	−0.01	0.00	0.00	0.00	−0.10***

Notes: CDS, clean dark spread; CSS, clean spark spread; SP, switching price; HOT, COLD, extreme temperatures days. *,**,***Significant at 0.05, 0.01 and 0.001 levels, respectively

	EUA	NAPII	NAPIII	OIL	GAS	COAL	CDS	CSS	SS	DAX	COLD
EUA											
NAPII	−0.04										
NAPIII	−0.01	−0.01									
OIL	0.07**	0.01	−0.01								
GAS	0.13***	−0.01	−0.04	0.11***							
COAL	−0.09**	0.01	0.00	0.07**	−0.04						
CDS	−0.03	0.01	−0.01	−0.06*	0.00	−0.05					
CSS	−0.03	0.01	0.00	−0.06*	−0.06*	−0.01	*0.95****				
SP	0.15***	−0.01	−0.04	0.07**	*0.93****	−0.33***	0.01	−0.05			
DAX	0.11***	0.04	−0.04	0.16***	0.02	−0.20***	0.04	0.03	0.10***		
COLD	−0.04	−0.01	0.02	−0.02	−0.01	−0.04	0.00	0.00	−0.01	−0.01	
HOT	−0.01	−0.01	−0.02	0.04	0.01	−0.04	0.03	0.03	0.02	0.03	−0.08**

Notes: CDS, clean dark spread; CSS, clean spark spread; SP, switching price; HOT, COLD, extreme temperatures days. *,**,***Significant at 0.05, 0.01 and 0.001 levels, respectively

Table V. Correlation table – Phase III

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4.6 Autocorrelation and heteroscedasticity tests

The EUA plots (Figure 1) detect the characteristic presence of heteroscedasticity which is confirmed by Box-Ljung and ARCH-LM tests in Table XI.

The ordinary least-squares (OLS) regression minimizes residuals and thereby produces the smallest standard errors. By definition, OLS regression gives equal weight to all observations, but when heteroscedasticity is present, the cases with larger disturbances become more influential. In this case, weighted least-squares regression would be more appropriate, as it down-weights those observations with larger disturbances. A more serious problem with heteroscedasticity is that the standard errors become biased. This leads to imprecise confidence intervals and incorrect conclusions about the significance of the regression coefficients. There are many approaches used here, e.g. using robust standard errors, weighted least-squares regression, transforming the dependent variable using one of the variance stabilizing transformations. A logarithmic transformation can be applied to highly skewed variables, while count variables can be transformed using a square root transformation. Overall, however, the violation of the homoscedasticity assumption must be quite severe in order to present a major problem given the robust nature of OLS regression. If we have an LR with generalized auto regressive conditional heteroscedasticity (GARCH) errors or auto regressive (AR) + GARCH model, we would still get consistent (although inefficient) estimates of the LR coefficients. Only when the true data generating process is ARIMA (ARIMA) + GARCH, ignoring the GARCH part will yield inconsistent estimates of the ARIMA coefficients and all the consequent troubles. That means the forecasts from an ARIMA models (with a neglected GARCH part) will likely be poor. We go ahead with linear models since the aim of the present study is to isolate the price determinants and determine their point forecasts.

5. Data modeling

After selecting the appropriate variables for modeling, we now attempt to build models using following seven modeling categories:

(1) LR;

(2) DLR;

(3) DLRLP;

(4) ARIMA;

(5) AE;

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Table VI. Correlation table – continuous series

	EUA	NAPI	NAPII	NAPIII	OIL	GAS	COAL	CDS	CSS	SS	DAX	COLD
EUA												
NAPI	0.03											
NAPII	0.01	0.00										
NAPIII	−0.01	0.00	−0.01									
OIL	0.13***	−0.01	0.02	0.00								
GAS	0.13***	−0.02	0.03	−0.02	0.11***							
COAL	−0.07***	0.01	0.00	0.00	0.03	0.00						
CDS	−0.02	−0.01	0.04*	−0.01	−0.03	0.03	−0.03					
CSS	−0.01	0.00	0.02	0.00	−0.04*	−0.10***	−0.01	*0.83****				
SP	0.10***	−0.01	0.02	−0.01	0.05**	*0.74****	−0.22***	0.02	−0.07***			
DAX	0.14***	0.00	0.01	−0.03	0.24***	0.05*	−0.14***	0.03	0.02	0.06**		
COLD	−0.03	−0.01	−0.03	0.02	0.03	0.00	0.01	−0.02	−0.02	−0.01	0.01	
HOT	−0.03	0.05*	−0.01	−0.01	0.01	0.00	−0.02	0.01	0.01	0.00	0.00	−0.10***

Notes: CDS, clean dark spread; CSS, clean spark spread; SP, switching price; HOT, COLD, extreme temperatures days. *,**,***Significant at 0.05, 0.01 and 0.001 levels, respectively

(6) AELP; and

(7) ARIMAX.

The first three are linear models and the last four are non-linear models. The data set is divided into two sets: estimation set and test set. The test set contains the last 30, 60, 90 and 120

Variable	RC1	RC2	RC5	RC10	RC8	RC4	RC7	RC9	RC6	RC3
NAPI	0.00	−0.01	0.00	0.00	*1.00*	0.02	0.00	−0.01	0.00	0.00
NAPII	0.02	0.02	0.00	0.01	0.00	−0.01	0.00	−0.01	*1.00*	0.01
NAPIII	0.00	−0.01	0.00	0.00	0.00	−0.01	*1.00*	0.01	0.00	−0.01
OIL	−0.02	0.05	0.02	*0.99*	0.00	0.00	0.00	0.01	0.01	0.12
GAS	−0.02	*0.94*	0.10	0.06	−0.01	0.00	−0.01	0.00	0.01	0.02
COAL	−0.01	−0.07	*0.99*	0.02	0.00	−0.01	0.00	0.00	0.00	−0.07
CDS	*0.96*	0.05	−0.01	−0.01	−0.01	0.00	0.00	−0.01	0.02	0.01
CSS	*0.96*	−0.08	0.00	−0.02	0.00	0.01	0.00	−0.01	0.00	0.00
SP	−0.01	*0.92*	−0.19	0.00	0.00	0.00	0.00	−0.01	0.01	0.02
DAX	0.02	0.03	−0.07	0.12	0.00	0.00	−0.01	0.00	0.01	*0.99*
COLD	−0.01	0.00	0.00	0.01	−0.01	−0.05	0.01	*1.00*	−0.01	0.00
HOT	0.01	0.00	−0.01	0.00	0.02	1.00	−0.01	−0.05	−0.01	0.00
SS loadings	1.84	1.75	1.03	1	1	1	1	1	1	1

Table VII. Principal components analysis – standardized loadings based upon correlation matrix (Continuous Series)

Variable	RC1	RC2	RC8	RC6	RC5	RC7	RC9	RC3	RC4
NAPI	−0.01	−0.01	0.01	−0.01	0.00	*1.00*	−0.01	0.04	−0.01
NAPII	0.05	0.02	0.03	0.03	0.01	−0.01	−0.04	−0.02	*1.00*
OIL	0.07	0.01	0.01	*0.99*	0.04	−0.01	0.01	0.00	0.03
GAS	*0.91*	−0.04	0.07	0.10	0.02	−0.02	−0.01	0.01	0.04
COAL	−0.06	0.00	*0.99*	0.01	−0.03	0.01	0.00	−0.01	0.03
CDS	0.06	*0.91*	0.03	0.03	0.04	−0.03	0.00	0.00	0.01
CSS	−0.12	*0.91*	−0.03	−0.02	−0.02	0.02	−0.03	0.00	0.01
SP	*0.91*	−0.01	−0.14	−0.02	0.00	0.00	0.00	0.00	0.02
DAX	0.02	0.01	−0.03	0.04	*1.00*	0.00	−0.02	−0.02	0.01
COLD	0.00	−0.03	0.00	0.01	−0.02	−0.01	*1.00*	−0.06	−0.04
HOT	0.01	0.00	−0.01	0.00	−0.02	0.04	−0.06	*1.00*	−0.02
SS loadings	1.68	1.66	1.01	1	1	1	1	1	1

Table VIII. Principal components analysis – standardized loadings based upon correlation matrix (Phase I)

Variable	RC1	RC2	RC5	RC10	RC8	RC7	RC4	RC9	RC6	RC3
NAPI	−0.01	0.00	0.00	−0.01	*1.00*	0.00	0.03	−0.01	0.00	0.00
NAPII	0.02	0.03	0.00	0.01	0.00	−0.01	−0.01	−0.02	*1.00*	0.01
NAPIII	−0.01	0.00	0.00	0.01	0.00	*1.00*	−0.01	0.01	−0.01	−0.03
OIL	0.05	−0.01	0.02	*0.99*	−0.01	0.01	0.00	0.02	0.01	0.14
GAS	*0.94*	−0.03	0.10	0.06	−0.01	−0.01	0.00	0.00	0.01	0.03
COAL	−0.07	−0.01	*0.99*	0.02	0.00	0.00	−0.01	0.01	0.00	−0.09
CDS	0.06	*0.92*	−0.01	0.00	−0.01	0.00	0.00	0.00	0.02	0.02
CSS	−0.11	*0.91*	−0.01	−0.01	0.01	0.00	0.00	−0.02	0.01	−0.01
SP	*0.92*	−0.02	−0.20	0.00	0.00	0.00	0.00	−0.01	0.01	0.02
DAX	0.04	0.01	−0.09	0.14	0.00	−0.03	0.00	0.01	0.01	*0.98*
COLD	−0.01	−0.02	0.01	0.02	−0.01	0.01	−0.05	*1.00*	−0.02	0.01
HOT	0.00	0.00	−0.01	0.00	0.03	−0.01	*1.00*	−0.05	−0.01	0.00
SS loadings	1.75	1.69	1.03	1	1	1	1	1	1	1

Table IX. Principal components analysis – standardized loadings based upon correlation matrix (Phase II)

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observations for EUA Phases I, II, III and continuous series. The estimation set contains remaining observations. In each category, the iterative modeling technique and intra-category comparison based on defined metrics such as adjusted R^2 and BIC are used to arrive at the best models in that category (modeled on estimation set) giving the significant predictive variables and/or the lags of dependent as well as predictive variables for EUA Phases I, II, III and continuous series. Afterwards, inter-category comparison is made between both estimation and test sets using the metric RMSE to decide the overall best models for EUA Phases I, II, III and continuous series giving the significant predictive variables and/or the lags of dependent as well as predictive variables, as the case may be.

5.1 LR

We employ the LR model as follows:

$$EUA_t = \alpha + \beta_1 RC1_t + \beta_2 RC2_t + \beta_3 NAPI_t + \beta_4 NAPII_t + \beta_5 NAPIII_t + \beta_6 brent_t + \beta_7 coal_t + \beta_8 DAX_t + \beta_9 TempCold_t + \beta_{10} TempHot_t + \epsilon_t \quad (5)$$

where EUA_t is the EUA return; α is the constant term; β_1 and β_2 are the parameters capturing the influence of the latent variables RC1 and RC2; β_3, β_4, β_5, β_9 and β_{10} are the parameters capturing the influence of dummy variables NAPI, NAPII and NAPIII, TempCold and TempHot, respectively; β_6, β_7 and β_8 are the parameters capturing the influence of variables brent, coal and DAX, respectively, and ε is the error term. The model is evaluated and insignificant predictors are removed from the model and the final models for EUA Phase I, EUA Phase II, EUA Phase III and the EUA continuous series are depicted in Equations (6), (7), (8) and (9), respectively:

$$EUA_t = \alpha + \beta_6 brent_t + \epsilon_t \quad (6)$$

Table X. Principal components analysis – standardized loadings based upon correlation matrix (Phase III)

Variable	RC2	RC1	RC4	RC3	RC8	RC5	RC6	RC7	RC9
NAPII	0.01	−0.01	0.00	0.02	−0.01	0.00	*1.00*	0.00	0.00
NAPIII	0.00	−0.02	0.00	−0.02	0.01	−0.01	0.00	*1.00*	0.00
OIL	−0.04	0.06	0.04	0.08	−0.01	0.02	0.00	0.00	*0.99*
GAS	−0.01	*0.99*	0.07	−0.01	0.00	0.00	0.00	−0.02	0.05
COAL	−0.02	−0.11	0.99	−0.10	−0.02	−0.02	0.00	0.00	0.04
CDS	*0.99*	0.02	−0.02	0.02	0.00	0.01	0.01	0.00	−0.02
CSS	*0.99*	−0.04	0.00	0.01	0.00	0.01	0.00	0.00	−0.02
SP	−0.01	*0.96*	−0.23	0.05	−0.01	0.01	0.00	−0.01	0.02
DAX	0.02	0.03	−0.10	*0.99*	0.00	0.01	0.02	−0.02	0.08
COLD	0.00	−0.01	−0.02	0.00	*1.00*	−0.04	−0.01	0.01	−0.01
HOT	0.02	0.01	−0.02	0.01	−0.04	*1.00*	0.00	−0.01	0.02
SS loadings	1.98	1.93	1.05	1	1	1	1	1	1

Table XI. Autocorrelation and heteroscedasticity tests for EUA returns

	EUA returns Phase I	EUA returns Phase II	EUA returns Phase III	EUA returns continuous series
Box-Ljung χ^2	216.53***	275.91***	64.45***	168.52***
DoF	12	12	12	12
p-value	< 0.001	< 0.001	< 0.001	< 0.001
ARCH-LM χ^2	308.43***	147.02***	42.99	104.36***
DoF	12	12	12	12
p-value	< 0.001	< 0.001	< 0.001	< 0.001

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$$EUA_t = \alpha + \beta_2 RC1_t + \beta_6 brent_t + \beta_7 coal_t + \beta_8 DAX_t + \epsilon_t \quad (7)$$

$$EUA_t = \alpha + \beta_2 RC2_t + \beta_6 brent_t + \beta_7 coal_t + \beta_8 DAX_t + \epsilon_t \quad (8)$$

$$EUA_t = \alpha + \beta_2 RC2_t + \beta_6 brent_t + \beta_7 coal_t + \beta_8 DAX_t + \epsilon_t \quad (9)$$

As seen from Table XII, Brent is significant in predicting the EUA returns for Phase I, RC1; Brent coal and DAX are significant in predicting EUA returns for Phase II and RC2; and Brent coal and DAX are significant in predicting EUA returns for Phase III and continuous series. So effectively, the significant predictors are RC1, RC2, Brent, coal and DAX. CDS and CSS have already loaded high on RC2 (RC1 for continuous series) so RC2 (RC1 for continuous series) has indirectly captured all the effect of gas returns also and that is why gas returns have not turned out to be a significant predictor. We also fail to detect the effects of news announcements about the allocation methodologies on the price of carbon. The R^2, adjusted R^2

	Estimate	SE	t value	Pr(> |t|)
Phase I (2005-2007)				
brent	0.7370	0.2409	3.059	0.002**
Model fitness criteria				
R^2	0.0136	Adjusted R^2	0.0121	
F(1,679)	9.36	p-value	0.002	
RMSE	0.10561	MAE	0.04549	
Phase II (2008-2012)				
RC1	0.0037	0.0006	6.204	< 0.001***
brent	0.1635	0.0307	5.329	< 0.001***
coal	−0.0868	0.0390	−2.228	0.026*
dax	0.2603	0.0422	6.173	< 0.001***
Model fitness criteria				
R^2	0.0713	Adjusted R^2	0.0694	
F(4,1969)	37.76	p-value	< 0.001	
RMSE	0.02630	MAE	0.01798	
Phase III (2013-2015)				
RC2	0.0041	0.0009	4.329	< 0.001***
brent	0.1256	0.0610	2.062	0.039*
coal	−0.1880	0.0913	−2.059	0.039*
DAX	0.2363	0.0776	3.045	0.002**
Model fitness criteria				
R^2	0.0353	Adjusted R^2	0.0322	
F(4,1257)	11.5	p-value	< 0.001	
RMSE	0.03350	MAE	0.02116	
Continuous (2005-2015)				
RC2	0.0033	0.0006	5.577	< 0.001***
Brent	0.1592	0.0313	5.081	< 0.001***
coal	−0.1049	0.0414	−2.536	0.011*
DAX	0.2346	0.0438	5.352	< 0.001***
Model fitness criteria				
R^2	0.0443	Adjusted R^2	0.0429	
F(4,2619)	30.39	p-value	< 0.001	
RMSE	0.02992	MAE	0.01942	

Note: *,**,***Significant at 0.05, 0.01 and 0.001 levels, respectively

Table XII. Linear regression model for EUA

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are too low to be interesting as a predicting model but the significant predictors are still relevant in predicting the change in EUA returns.

Since DLRLP models are the best models found by this study out of all LR class models, namely LR, DLR and DLRLP models, the diagnostics of LR class models are depicted for DLRLP models only in Figure 4.

5.2 Dynamic LR

We now modify the regression model by including the lags (maximum 20 lags are tried) of the dependent variable as predictors. The general model is given as follows:

$$EUA_t = \alpha + \sum_{i=1}^{20} \phi_i EUA_{t-i} + \beta_1 RC1_t + \beta_2 RC2_t + \beta_3 NAPI_t + \beta_4 NAPII_t + \beta_5 NAPIII_t + \beta_6 brent_t + \beta_7 coal_t + \beta_8 DAX_t + \beta_9 TempCold_t + \beta_{10} TempHot_t + \epsilon_t \quad (10)$$

where α is the intercept, EUA_{t-i} is ith lag of EUA return, ϕ_i is the parameter capturing the influence of the AR(i) process, β_1 and β_2 are the parameters capturing the influence of the latent variables RC1 and RC2, β_3, β_4, β_5, β_9 and β_{10} are the parameters capturing the influence of dummy variables NAPI, NAPII and NAPIII, TempCold and TempHot, respectively, β_6, β_7 and β_8 are the parameters capturing the influence of variables Brent, coal and DAX, respectively and ϵ_t is the error term. The models are evaluated and insignificant predictors are removed from the model. The final predictors are given in Table XIII and the final models for EUA Phase I, EUA Phase II, EUA Phase III and the EUA continuous series are depicted in Equations (11), (12), (13) and 14, respectively:

$$EUA_t = \alpha + \phi_2 EUA_{t-1} + \phi_{10} EUA_{t-10} + \phi_{11} EUA_{t-11} + \phi_{12} EUA_{t-12} + \phi_{13} EUA_{t-13} + \phi_{16} EUA_{t-16} + \phi_{17} EUA_{t-17} + \phi_{18} EUA_{t-18} + \beta_6 brent_t + \epsilon_t \quad (11)$$

$$EUA_t = \alpha + \phi_2 EUA_{t-1} + \phi_6 EUA_{t-6} + \phi_7 EUA_{t-7} + \phi_{11} EUA_{t-11} + \phi_{18} EUA_{t-18} + \beta_7 RC1_t + \beta_6 brent_t + \beta_7 coal_t + \beta_8 DAX_t + \epsilon_t \quad (12)$$

$$EUA_t = \alpha + \phi_{12} EUA_{t-12} + \phi_{17} EUA_{t-17} + \beta_2 RC2_t + \beta_8 DAX_t + \epsilon_t \quad (13)$$

$$EUA_t = \alpha + \phi_1 EUA_{t-1} + \phi_2 EUA_{t-2} + \phi_4 EUA_{t-4} + \phi_6 EUA_{t-6} + \phi_{12} EUA_{t-12} + \phi_{13} EUA_{t-13} + \phi_{17} EUA_{t-17} + \phi_{18} EUA_{t-18} + \phi_{19} EUA_{t-19} + \beta_2 RC2_t + \beta_6 brent_t + \beta_7 coal_t + \beta_8 DAX_t + \epsilon_t. \quad (14)$$

5.3 DLRLP

We now further modify the regression model by including the current value and the lags of the independent variables also as predictors. The models are evaluated and insignificant predictors are removed from the model. The final predictors are given in Table XIV and the final models for EUA Phase I, EUA Phase II, EUA Phase III and the EUA continuous series are depicted in Equations (15), (16), (17) and (18), respectively:

$$EUA_t = \alpha + \phi_2 EUA_{t-1} + \phi_{10} EUA_{t-10} + \phi_{11} EUA_{t-11} + \phi_{12} EUA_{t-12} + \phi_{13} EUA_{t-13} + \phi_{16} EUA_{t-16} + \phi_{17} EUA_{t-17} + \phi_{18} EUA_{t-18} + \beta_1 RC1_{t-11} + \beta_2 RC1_{t-12} + \beta_3 RC2_{t-12} + \beta_4 RC2_{t-20} + \beta_5 napI_{t-6} + \beta_6 napI_{t-10} + \beta_7 brent_t + \beta_8 coal_{t-18} + \beta_9 coal_{t-19} + \beta_{10} dax_{t-8} + \epsilon_t \quad (15)$$

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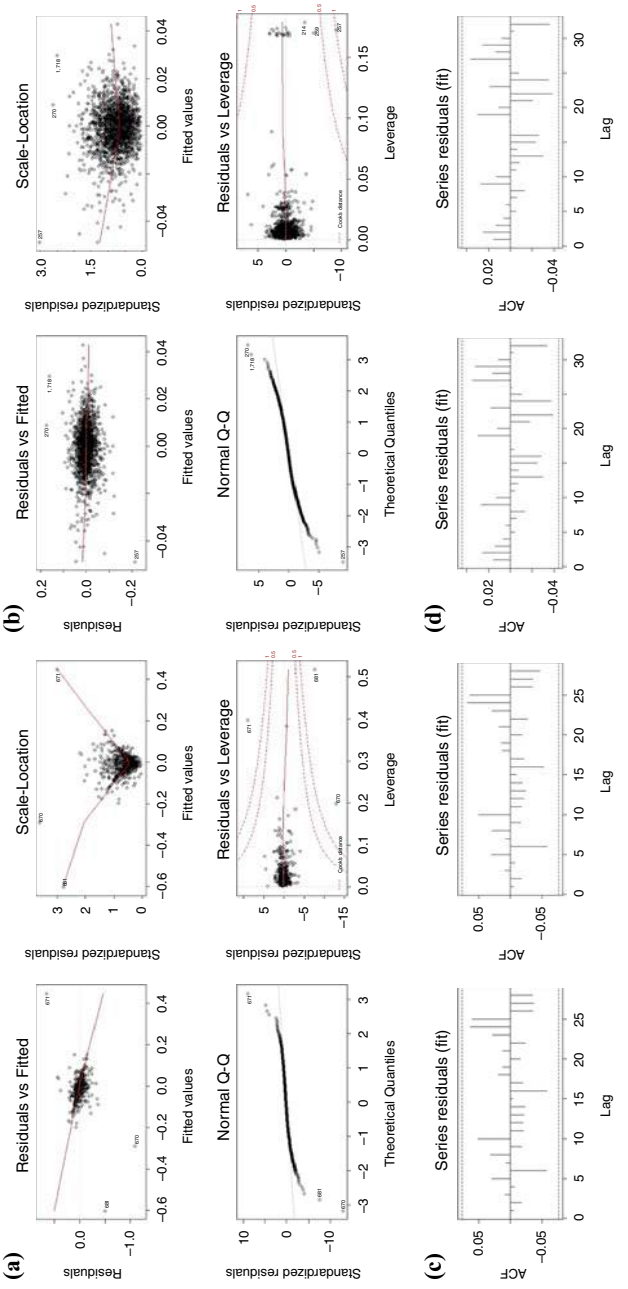

Figure 4.
Dynamic linear regression with lagged predictors: plots for the residuals

(continued)

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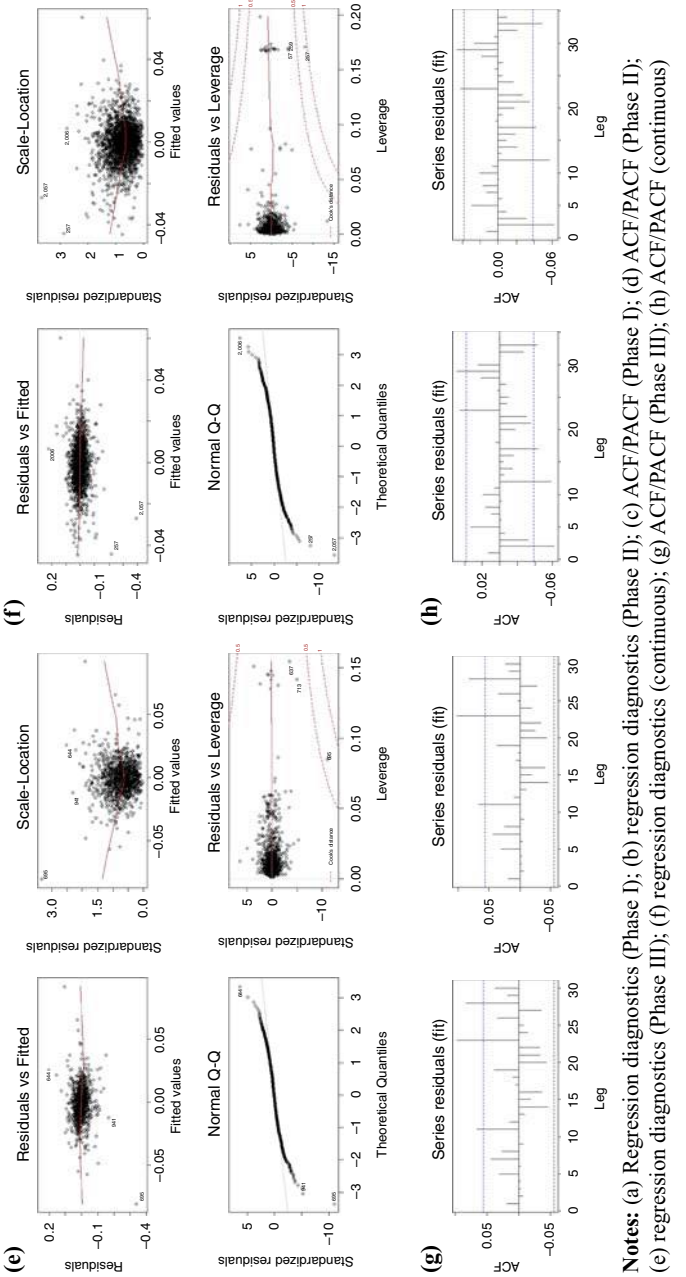

Notes: (a) Regression diagnostics (Phase I); (b) regression diagnostics (Phase II); (c) ACF/PACF (Phase I); (d) ACF/PACF (Phase II); (e) regression diagnostics (Phase III); (f) regression diagnostics (continuous); (g) ACF/PACF (Phase III); (h) ACF/PACF (continuous)

Figure 4.

	Estimate	SE	t value	Pr(> |t|)
Phase I (2005-2007)				
eua Lag 1	−0.1556	0.0396	−3.92	< 0.001***
eua Lag 10	−0.1099	0.0402	−2.73	0.006**
eua Lag 11	0.2537	0.0448	5.66	< 0.001***
eua Lag 12	0.1432	0.0554	2.58	0.009**
eua Lag 13	−0.1505	0.0567	−2.65	0.008**
eua Lag 16	−0.1164	0.0569	−2.04	0.041*
eua Lag 17	−0.1805	0.0568	−3.17	0.001**
eua Lag 18	0.1607	0.0571	2.81	0.005**
brent	0.7007	0.2302	3.04	0.002**
Model fitness criteria				
R^2	0.1487	Adjusted R^2	0.1369	
F(9,653)	12.67	p-value	< 0.001	
RMSE	0.09936	MAE	0.05136	
Phase II (2008-2012)				
eua Lag 1	0.0784	0.0217	3.60	< 0.001***
eua Lag 6	−0.0822	0.0217	−3.78	< 0.001***
eua Lag 7	0.0628	0.0217	2.88	0.004**
eua Lag 11	−0.0473	0.0217	−2.17	0.029*
eua Lag 18	−0.0626	0.0218	−2.87	0.004**
RC1	0.0036	0.0005	6.09	< 0.001***
brent	0.1779	0.0305	5.82	< 0.001***
coal	−0.0852	0.0389	−2.18	0.028*
dax	0.2670	0.0418	6.37	< 0.001***
Model fitness criteria				
R^2	0.0930	Adjusted R^2	0.0889	
F(9,1946)	22.2	p-value	< 0.001	
RMSE	0.02603	MAE	0.0179	
Phase III (2013-2015)				
eua Lag 12	−0.0714	0.0277	−2.579	0.010*
eua Lag 17	−0.1170	0.0277	−4.226	< 0.001***
RC2	0.0045	0.0009	4.745	< 0.001***
dax	0.2969	0.0751	3.949	< 0.001***
Model fitness criteria				
R^2	0.0494	Adjusted R^2	0.0463	
F(4,1240)	16.11	p-value	< 0.001	
RMSE	0.03344	MAE	0.02139	
Continuous (2005-2015)				
eua Lag 1	0.0578	0.0190	3.04	0.002**
eua Lag 2	−0.0634	0.0189	−3.34	< 0.001***
eua Lag 4	0.0685	0.0190	3.60	< 0.001***
eua Lag 6	−0.0438	0.0190	−2.29	0.021*
eua Lag 12	−0.0505	0.0189	−2.66	0.007**
eua Lag 13	−0.0817	0.0189	−4.30	< 0.001***
eua Lag 17	−0.0397	0.0190	−2.08	0.036*
eua Lag 18	−0.0729	0.0189	−3.84	< 0.001***
eua Lag 19	0.0619	0.0191	3.24	0.001**
RC2	0.0034	0.0005	5.89	< 0.001***
brent	0.1644	0.0309	5.32	< 0.001***
coal	−0.1059	0.0409	−2.58	0.009**
dax	0.2421	0.0431	5.60	< 0.001***
Model fitness criteria				
R^2	0.08251	Adjusted R^2	0.07791	
F(13,2591)	17.92	p-value	< 0.001	
RMSE	0.02936	MAE	0.01935	

Note: *,**,***Significant at 0.05, 0.01 and 0.001 levels, respectively

Table XIII. Dynamic linear regression model for EUA

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	Estimate	SE	t value	Pr(> |t|)
Phase I (2005-2007)				
eua Lag 1	−0.1310	0.0382	−3.42	< 0.001***
eua Lag 10	−0.1025	0.0379	−2.69	0.007**
eua Lag 11	0.2244	0.0434	5.17	< 0.001***
eua Lag 12	0.1383	0.0524	2.63	0.008**
eua Lag 13	−0.1905	0.0532	−3.57	< 0.001***
eua Lag 17	−0.1692	0.0536	−3.15	0.001**
eua Lag 18	0.1508	0.0540	2.78	0.005**
RC1 Lag 11	0.0082	0.0037	2.19	0.028*
RC1 Lag 12	−0.0117	0.0037	−3.14	0.001**
RC2 Lag 12	−0.0079	0.0036	−2.18	0.028*
RC2 Lag 20	−0.0087	0.0036	−2.36	0.018*
napI Lag 6	−0.0842	0.0248	−3.38	< 0.001***
napI Lag 10	−0.0575	0.0248	−2.31	0.020*
brent	0.8002	0.2175	3.67	< 0.001***
coal Lag 18	−1.9313	0.3587	−5.38	< 0.001***
coal Lag 19	1.5833	0.3637	4.35	< 0.001***
dax Lag 8	−1.1386	0.3968	−2.86	0.004**
Model fitness criteria				
R^2	0.2609	Adjusted R^2	0.2413	
F(17,643)	13.35	p-value	< 0.001	
RMSE	0.09567	MAE	0.05296	
Phase II (2008-2012)				
eua Lag 1	0.0689	0.0215	3.19	0.001**
eua Lag 6	−0.0865	0.0214	−4.03	< 0.001***
eua Lag 7	0.0675	0.0216	3.12	0.001**
eua Lag 11	−0.0425	0.0215	−1.97	0.048*
eua Lag 18	−0.0617	0.0216	−2.84	0.004**
RC1 Lag 0	0.0037	0.0005	6.37	< 0.001***
RC1 Lag 3	−0.0015	0.0005	−2.69	0.007**
napI Lag 1	−0.0273	0.0106	−2.57	0.010*
napI Lag 11	−0.0368	0.0105	−3.48	< 0.001***
napII Lag 9	−0.0282	0.0105	−2.66	0.007**
napII Lag 14	0.0224	0.0105	2.12	0.034*
brent	0.1714	0.0301	5.68	< 0.001***
coal Lag 1	−0.0969	0.0377	−2.56	0.010*
dax	0.2827	0.0408	6.92	< 0.001***
tempcold Lag 15	0.0026	0.0012	2.10	0.035*
temphot Lag 18	−0.0090	0.0040	−2.23	0.025*
Model fitness criteria				
R^2	0.1156	Adjusted R^2	0.1083	
F(16,1939)	15.83	p-value	< 0.001	
RMSE	0.02571	MAE	0.01807	
Phase III (2013-2015)				
eua Lag 2	−0.0966	0.0269	−3.58	< 0.001***
eua Lag 3	−0.0848	0.0268	−3.16	0.001**
eua Lag 4	0.0839	0.0271	3.09	0.002**
eua Lag 12	−0.0698	0.0268	−2.60	0.009**
eua Lag 13	−0.0876	0.0268	−3.25	0.001**
eua Lag 17	−0.0939	0.0272	−3.44	< 0.001***
eua Lag 18	−0.0633	0.0266	−2.37	0.017*

(*continued*)

Table XIV. Dynamic linear regression with lagged predictors model for EUA

	Estimate	SE	t value	Pr(> |t|)
RC1 Lag 4	−0.0026	0.0009	−2.90	0.003**
RC1 Lag 15	−0.0044	0.0009	−4.85	< 0.001***
RC2 Lag 0	0.0048	0.0009	5.24	< 0.001***
RC2 Lag 19	−0.0024	0.0009	−2.71	0.006**
napIII Lag 11	−0.0155	0.0066	−2.32	0.020*
napIII Lag 20	−0.0166	0.0066	−2.49	0.012*
coal Lag 11	0.2536	0.0854	2.96	0.003**
dax	0.2923	0.0726	4.02	< 0.001***
dax Lag 6	−0.1769	0.0728	−2.42	0.015*
dax Lag 17	0.1551	0.0739	2.09	0.036*
Model fitness criteria				
R^2	0.1361	Adjusted R^2	0.1241	
F(17,1224)	11.34	p-value	< 0.001	
RMSE	0.03191	MAE	0.02129	
Continuous (2005-2015)				
eua Lag 1	0.0494	0.0189	2.61	0.009**
eua Lag 4	0.0809	0.0189	4.28	< 0.001***
eua Lag 6	−0.0438	0.0189	−2.31	0.020*
eua Lag 13	−0.0903	0.0188	−4.78	< 0.001***
eua Lag 18	−0.0711	0.0189	−3.75	< 0.001***
eua Lag 19	0.0660	0.0189	3.48	< 0.001***
RC2 Lag 0	0.0035	0.0005	6.17	< 0.001***
RC2 Lag 3	−0.0015	0.0005	−2.61	0.008**
napI Lag 9	−0.0332	0.0120	−2.76	0.005**
napI Lag 11	−0.0301	0.0120	−2.50	0.012*
napII Lag 15	0.0290	0.0120	2.40	0.016*
napII Lag 18	−0.0355	0.0120	−2.95	0.003**
brent	0.1660	0.0308	5.38	< 0.001***
dax	0.2564	0.0426	6.01	< 0.001***
Model fitness criteria				
R^2	0.0844	Adjusted R^2	0.0795	
F(14,2590)	17.07	p-value	< 0.001	
RMSE	0.02933	MAE	0.01950	

Note: *,**,***Significant at 0.05, 0.01 and 0.001 levels, respectively

Table XIV.

$$
\begin{aligned}
EUA_t = {} & \alpha + \phi_2 EUA_{t-1} + \phi_6 EUA_{t-6} + \phi_7 EUA_{t-7} + \phi_{11} EUA_{t-11} + \phi_{18} EUA_{t-18} \\
& + \beta_1 RC1_t + \beta_2 RC1_{t-3} + \beta_3 napI_{t-1} + \beta_4 napI_{t-11} + \beta_5 napII_{t-9} \\
& + \beta_6 napII_{t-14} + \beta_7 brent_t + \beta_8 coal_{t-1} + \beta_9 dax_t + \beta_{10} tempcold_{t-15} \\
& + \beta_{11} temphot_{t-18} + \epsilon_t
\end{aligned} \tag{16}
$$

$$
\begin{aligned}
EUA_t = {} & \alpha + \phi_2 EUA_{t-2} + \phi_3 EUA_{t-3} + \phi_4 EUA_{t-4} + \phi_{12} EUA_{t-12} + \phi_{13} EUA_{t-13} \\
& + \phi_{17} EUA_{t-17} + \phi_{18} EUA_{t-18} + \beta_1 RC1_{t-4} + \beta_2 RC1_{t-15} + \beta_3 RC2_t \\
& + \beta_4 RC2_{t-19} + \beta_4 napIII_{t-11} + \beta_5 napIII_{t-20} + \beta_6 coal_{t-11} + \beta_7 dax_t \\
& + \beta_8 dax_{t-6} + \beta_9 dax_{t-17} + \epsilon_t
\end{aligned} \tag{17}
$$

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$$EUA_t = \alpha + \phi_1 EUA_{t-1} + \phi_4 EUA_{t-4} + \phi_6 EUA_{t-6} + \phi_{13} EUA_{t-13} + \phi_{18} EUA_{t-18} + \phi_{19} EUA_{t-19} + \beta_1 RC2_t + \beta_2 RC2_{t-3} + \beta_3 napI_{t-9} + \beta_4 napI_{t-1} + \beta_5 napI_{t-15} + \beta_6 napI_{t-18} + \beta_7 brent_t + \beta_{10} dax_t + \epsilon_t \quad (18)$$

where α is the intercept, EUA_{t-i} is ith lag of EUA return, ϕ_i is the parameter capturing the influence of the AR(i) process, β_i's are the parameters capturing the influence of the various lags of latent variables RC1 and RC2, and other dependent variables NAPI, NAPII, NAPIII, TempCold, TempHot, brent, coal and DAX and ϵ_t is the error term.

The diagnostic plots (Figure 4(a), (b), (e) and (f)) show that the models are misspecified and the assumption $E(\varepsilon|X=0)$ does not hold true. Q-Q plots also show departure from normality. The "residuals vs fitted" and "scale-location" charts are essentially the same, and show if there is a trend to the residuals. OLS models require that the residuals be "identically and independently distributed," that their distribution does not change substantially for different values of independent variables. All the charts are moderately satisfactory in this regard. If this assumption is not fully met, the β estimates are still good, but the t-statistics, and corresponding p-values, are troublesome. Another assumption is that the errors are approximately normally distributed, which is what the Q-Q plot shows. Again, the plots are moderately satisfactory in this regard. The consequences of this assumption not being fully met are the same, i.e. βs are good and ts are troublesome.

The ACF and PACF plots (Figure 4(c), (d), (g), (h)), being both sinusoidal, give the mixed signal for the orders of AR and moving averages (MA) processes. Figure 5 gives a depiction of both original and fitted values for all three phases and the continuous series for comparison purposes.

5.4 ARIMA model

In a multiple regression model, we forecast the variable using a linear combination of predictors. In an AR model, we forecast the variable using a linear combination of past

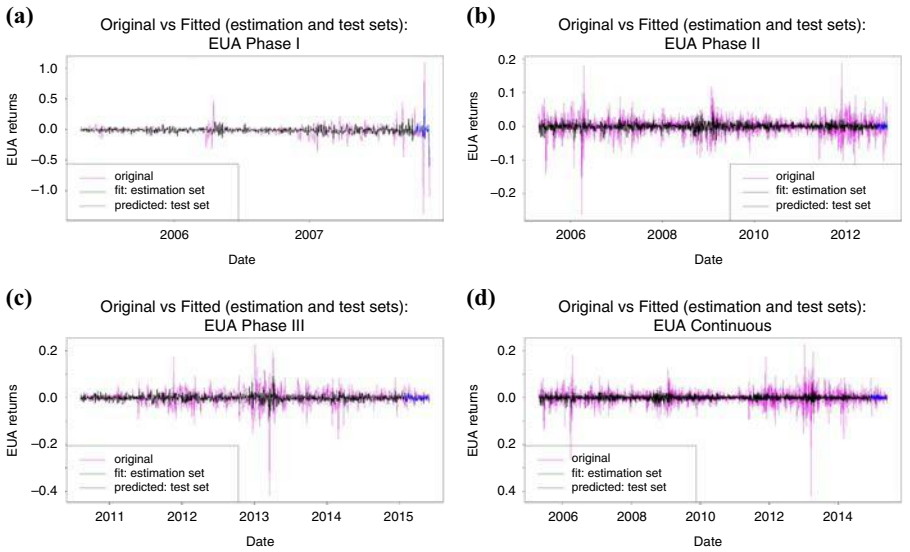

Figure 5. Dynamic linear regression with lagged predictors model for EUA: original vs fitted values

Notes: (a) EUA Phase I; (b) EUA Phase II; (c) EUA Phase III; (d) EUA continuous series

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values of the same variable. So it is a regression of the variable against itself. Rather than using past values of the forecast variable, a MA model uses past forecast errors as predictors. If we combine differencing with AR and a MA model, we get a non-seasonal ARIMA model ("integration" here is the reverse of differencing). The ARIMA (p, d, q) model can be written as follows:

$$y'_t = c + \phi_1 y'_{t-1} + \cdots + \phi_p y'_{t-p} + \theta_1 e_{t-1} + \cdots + \theta_q e_{t-q} + \epsilon_t. \tag{19}$$

The predictors here are both lagged values of y_t and lagged error terms. Here p is the order or AR part, d is degree of differencing done, and q is the order of MA part.

Good models are obtained by minimizing either the Akaike's information criterion (AIC), corrected Akaike's information criterion (AIC_c) or BIC. We prefer to use the BIC because BIC gives the parsimonious model.

All the variables used in the model exhibit stationarity (being already differenced). The EUA return series does not show any seasonality in ACF plots and *R* function *tbats* also resulted in seasonality as false. The iterative procedure is adopted to find out the various models with significant lags as predictors. All possible models (based on ACF/PACF plots of EUA) are examined to verify the white noise character of residuals using ACF plots of residuals and the Portmanteau test of the residuals. Best model is then selected out of all verified (for the white noise character of residuals) models, which gives the least BIC. The model thus obtained is given in Table XV. We found MA1, MA11 and MA13 as significant predictors for EUA Phase I; AR1, AR6 and AR7 as significant predictors for EUA Phase II; AR2, AR4, AR13, AR17, MA3, MA11, MA12 and MA23 as significant predictors for EUA Phase III and AR1, AR2, AR4, AR6, AR13, AR17, MA12 and MA17 as significant predictors for EUA continuous series. The ARs and MAs, here, are auto regressive and moving average terms.

Since AE models are the best models found by this study out of all ARIMA class models, namely ARIMA, AE, AELP and ARIMAX, the diagnostics of ARIMA class are depicted for AE models only in Figures 6, 8, 10 and 12.

5.5 AE model

In simple LR, we consider the model (Hyndman and Athanasopoulos, 2013) as follows:

$$y_t = \beta_0 + \beta_1 x_{1,t} + \cdots + \beta_k x_{k,t} + \epsilon_t \tag{20}$$

where y_t is a linear function of the k predictor variables ($x_{1,\ t}, \ldots, x_{k,t}$), and ϵ_t is a uncorrelated error term, i.e. white noise. In regression with AE, we allow the errors from a regression to contain autocorrelation, i.e.:

$$y_t = \beta_0 + \beta_1 x_{1,t} + \ldots + \beta_k x_{k,t} + \eta_t \tag{21}$$

and η_t is assumed to follow an ARIMA model. For example, if η_t follows an ARIMA (1, 1, 1) model, we can write:

$$(1-\phi_1 B)(1-B)\eta_t = (1+\theta_1 B)\epsilon_t \tag{22}$$

where ϵ_t is the white noise. We now have two error terms here – the error from the regression model which is denoted as η_t and the error from the ARIMA model which is denoted as ϵ_t. Only the error from the ARIMA model, i.e. ϵ_t is assumed as the white noise.

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	Estimate	SE	t value	Pr(> |t|)
Phase I (2005-2007)				
ma1	−0.1786	0.0368	−4.85	< 0.001***
ma11	0.324717	0.0355	9.12	< 0.001***
ma13	−0.1894	0.0435	−4.34	< 0.001***
Model fitness criteria				
AIC_c	−1,185.66	BIC	−1,167.62	
RMSE	0.10058	MAE	0.04847	
Box-Ljung $\chi^2(33)$	32.86	p-value	0.474	
ΣAR	0	ΣMA	−0.0433	
Phase II (2008-2012)				
ar1	0.0722	0.0224	3.22	0.001**
ar6	−0.0808	0.0224	−3.60	< 0.001***
ar7	0.0624	0.0225	2.77	0.005**
Model fitness criteria				
AIC_c	−8,635.45	BIC	−8,613.12	
RMSE	0.02709	MAE	0.01891	
Box-Ljung $\chi^2(33)$	35.71	p-value	0.342	
ΣAR	0.0538	ΣMA	0	
Phase III (2013-2015)				
ar2	−0.1041	0.0278	−3.74	< 0.001***
ar4	0.0801	0.0279	2.86	0.004**
ar13	−0.0921	0.0276	−3.33	< 0.001***
ar17	−0.0964	0.02	−3.4660	< 0.001***
ma3	−0.0781	0.0285	−2.73	0.006**
ma11	0.0752	0.0294	2.55	0.010*
ma12	−0.0855	0.0282	−3.03	0.002**
ma23	0.0890	0.0274	3.24	0.001**
Model fitness criteria				
AIC_c	−5,015.31	BIC	−4,969.19	
RMSE	0.03292	MAE	0.02152	
Box-Ljung $\chi^2(28)$	29.79	p-value	0.373	
ΣAR	−0.2126	ΣMA	0.0006	
Continuous (2005-2015)				
ar1	0.0667	0.0167	3.98	< 0.001***
ar2	−0.0703	0.0171	−4.10	< 0.001***
ar4	0.0603	0.0176	3.42	< 0.001***
ar6	−0.0517	0.0173	−2.98	0.002**
ar13	−0.0755	0.0173	−4.35	< 0.001***
ar17	−0.4731	0.0724	−6.53	< 0.001***
ma12	−0.0591	0.0174	−3.39	< 0.001***
ma17	0.4517	0.0746	6.05	< 0.001***
Model fitness criteria				
AIC_c	−10,935.13	BIC	−10,882.34	
RMSE	0.03001	MAE	0.02005	
Box-Ljung $\chi^2(28)$	23.69	p-value	0.697	
ΣAR	−0.5436	ΣMA	0.3926	

Note: *,**,***Significant at 0.05, 0.01 and 0.001 levels, respectively

Table XV. ARIMA model for EUA

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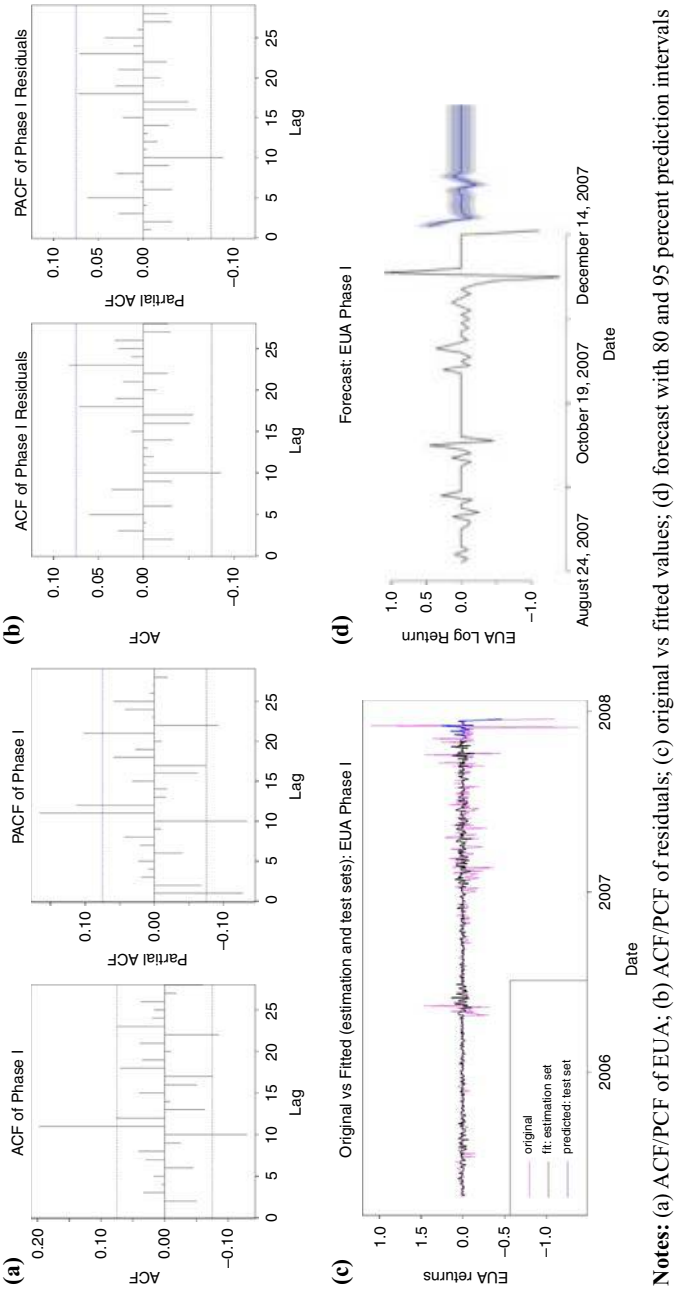

Notes: (a) ACF/PCF of EUA; (b) ACF/PCF of residuals; (c) original vs fitted values; (d) forecast with 80 and 95 percent prediction intervals

Figure 6.
ARIMA errors model for EUA Phase I

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Since all the variables in our models are differences and stationary, our model becomes:

$$y'_t = \beta_0 + \beta_1 x'_{1,t} + \ldots + \beta_k x'_{k,t} + \eta'_t \tag{23}$$

$$(1-\phi_1 B)(1-B)\eta'_t = (1+\theta_1 B)\epsilon_t \tag{24}$$

where $y'_t = y_t - y_{t-1}$, $x'_{t,i} = x_{t,i} - x_{t-1,i}$ and $\eta'_t = \eta_t - \eta_{t-1}$. This is a regression model in differences with ARMA errors. The full modeling procedure adopted is as follows:

(1) EUA series and all predictors are already differenced and stationary variables;

(2) the regression model is fitted with a proxy model with AR(2) errors;

(3) the errors (n_t) is calculated from the fitted regression model and an appropriate ARMA model for them is identified;

(4) re-fit the entire model using the new ARMA model for the errors; and

(5) the e_t series is checked for looking like white noise.

Similar to ARIMA modeling, the iterative procedure is again adopted to find out the various models with significant lags of AE as predictors. All possible models (based on ACF/PACF plots of EUA) are examined to verify the white noise character of residuals using ACF plots of residuals and the Portmanteau test of the residuals. Best model is then selected out of all verified (for the white noise character of residuals) models, which gives the least BIC. The model thus obtained is as follows.

Figures 6(b) and 7; 8(b) and 9; 10(b) and 11; and 12(b) and 13 show that the ACF of the residuals is a white noise (which is also confirmed from the Box-Ljung Test and the *p*-values of the Portmanteau test) of EUA Phases I, II, III and continuous, respectively. Figures 6(c), 8(c), 10(c) and 12(c) depict the accuracy of the model for the in-sample as well as out-of-sample observations. Figures 6(d), 8(d), 10(d) and 12(d) give the forecast from the model for the next 30 days after out-of-sample values.

Table XVI shows that all the AR coefficients and MA coefficients are significantly different from zero and neither the sum of AR coefficients nor the sum of MA coefficients is close to unity so there are no signs of unit roots. On the whole, this appears to be a good model. We found MA1, MA11, MA13 and Brent as significant predictors for EUA Phase I; AR1, AR6, AR7, AR18, RC1, Brent and DAX as significant predictors for EUA Phase II;

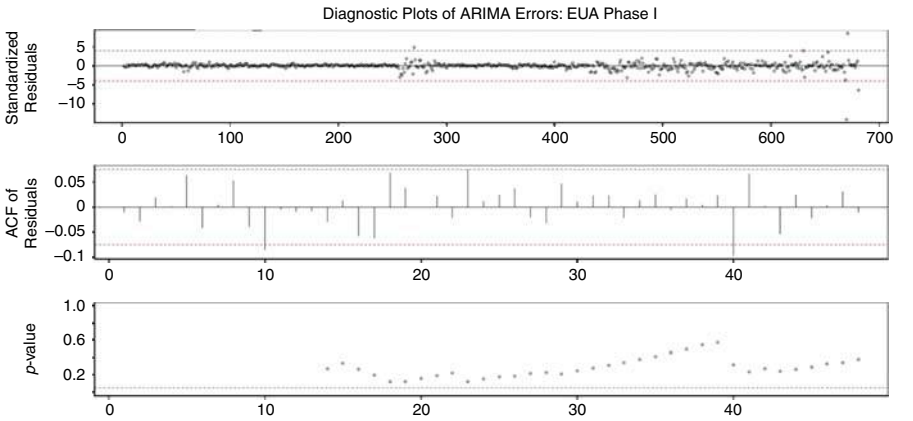

Figure 7.
ARIMA errors model for EUA Phase I: diagnostic plots: standardized residuals, ACF of residuals and the *p*-values of Portmanteau test

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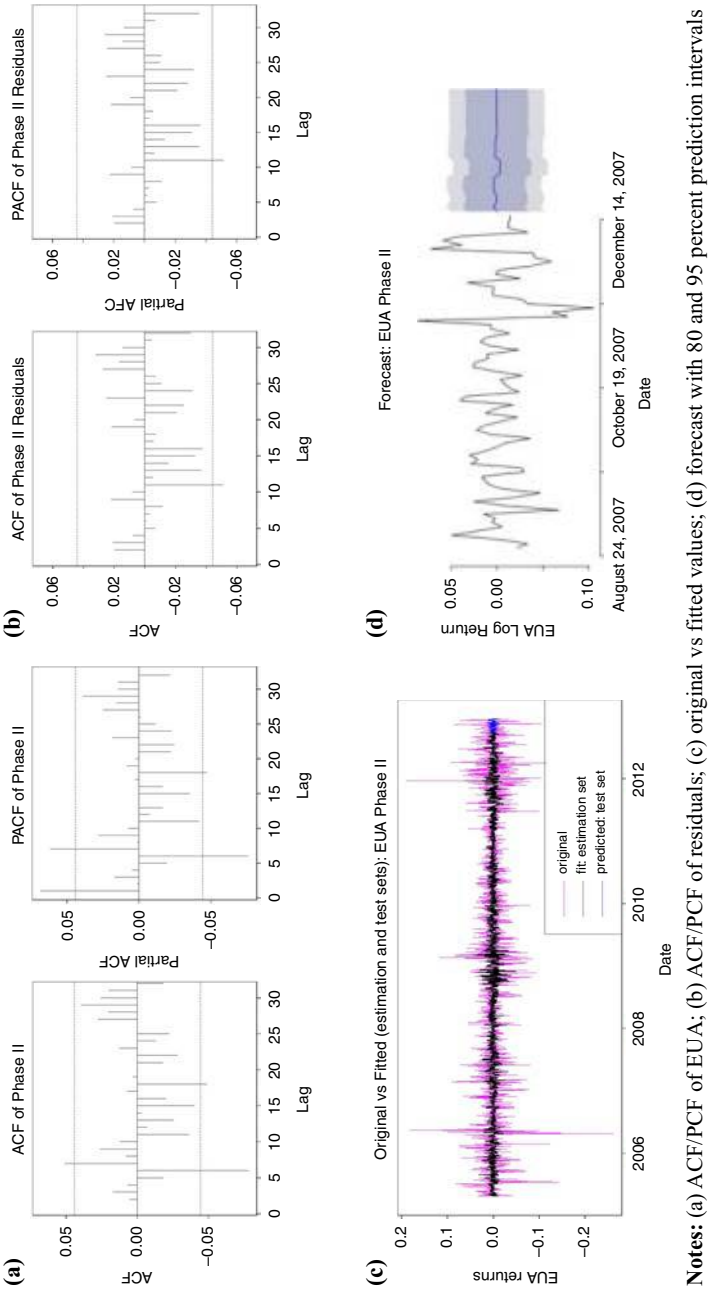

Notes: (a) ACF/PCF of EUA; (b) ACF/PCF of residuals; (c) original vs fitted values; (d) forecast with 80 and 95 percent prediction intervals

Figure 8.
ARIMA errors model for EUA Phase II

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AR2, AR4, AR13, AR17, MA11, MA12, MA23, RC2, Brent, coal and DAX as significant predictors for EUA Phase III; and AR1, AR2, AR4, AR6, AR13, AR17, MA12, MA17, RC2, Brent, coal, DAX and temphot as significant predictors for EUA continuous series.

5.6 AELP

This model is similar to the previous model with one more additional aspect that lagged predictors are also taken in here to predict the dependent variable. Similar to ARIMA modeling, the iterative procedure is again adopted to find out the various models with significant lags of AE and lagged predictors as the overall predictors. All possible models (based on ACF/PACF plots of EUA) are examined to verify the white noise character of residuals using ACF plots of residuals and the Portmanteau test of the residuals. Best model is then selected out of all verified (for the white noise character of residuals) models, which gives the least BIC. The model thus obtained is as follows.

Table XVII shows that all the AR coefficients, MA coefficients and the coefficients of lagged predictors are significantly different from zero and neither the sum of AR coefficients nor the sum of MA coefficients is close to unity so there are no signs of unit roots. On the whole, this appears to be a good model. We found MA1, MA11, MA13 and Brent as the significant predictors for EUA Phase I; AR1, AR6, AR7, AR18, RC1, Brent and DAX as the significant predictors for EUA Phase II; AR2, AR4, AR13, AR17, MA11, MA12, MA23, RC2, Brent, coal and DAX as significant predictors for EUA Phase III; and AR1, AR2, AR4, AR6, AR13, AR17, MA12, MA17, RC2, Brent, coal, DAX and temphot as significant predictors for EUA continuous series.

5.7 ARIMA with transfer function

Assume two time series denoted EUA_t and X_t, which are both stationary. Then, the transfer function model (TFM) can be written as follows:

$$EUA_t = \mathrm{C} + v(B)X_t + N_t \tag{25}$$

where EUA_t is the dependent variable, X_t is the independent variable, C is the constant term, N_t is the stochastic disturbance and $\nu(B)X_t$ is the transfer function (or impulse response function) which allows X to influence EUA via a distributed lag B as a backshift operator:

$$v(B)X_t = \left(v_0 + v_1 B + v_2 B^2 + \ldots\right)X_t \tag{26}$$

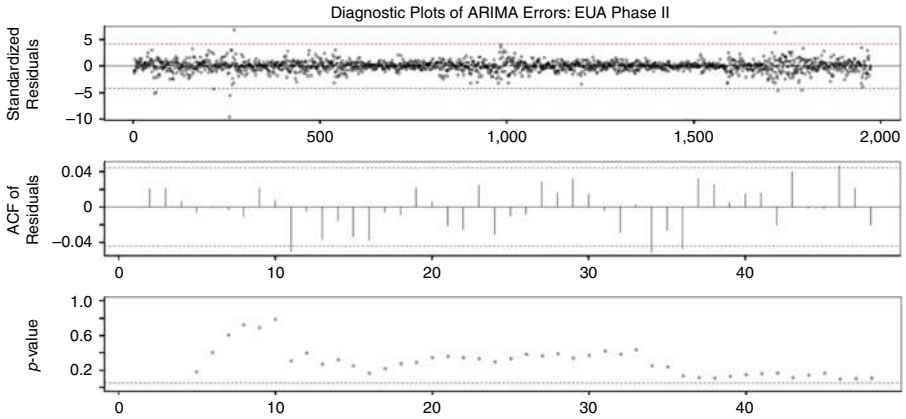

Figure 9.
ARIMA errors model for EUA Phase II: diagnostic plots: standardized residuals, ACF of residuals, and the *p*-values of Portmanteau test

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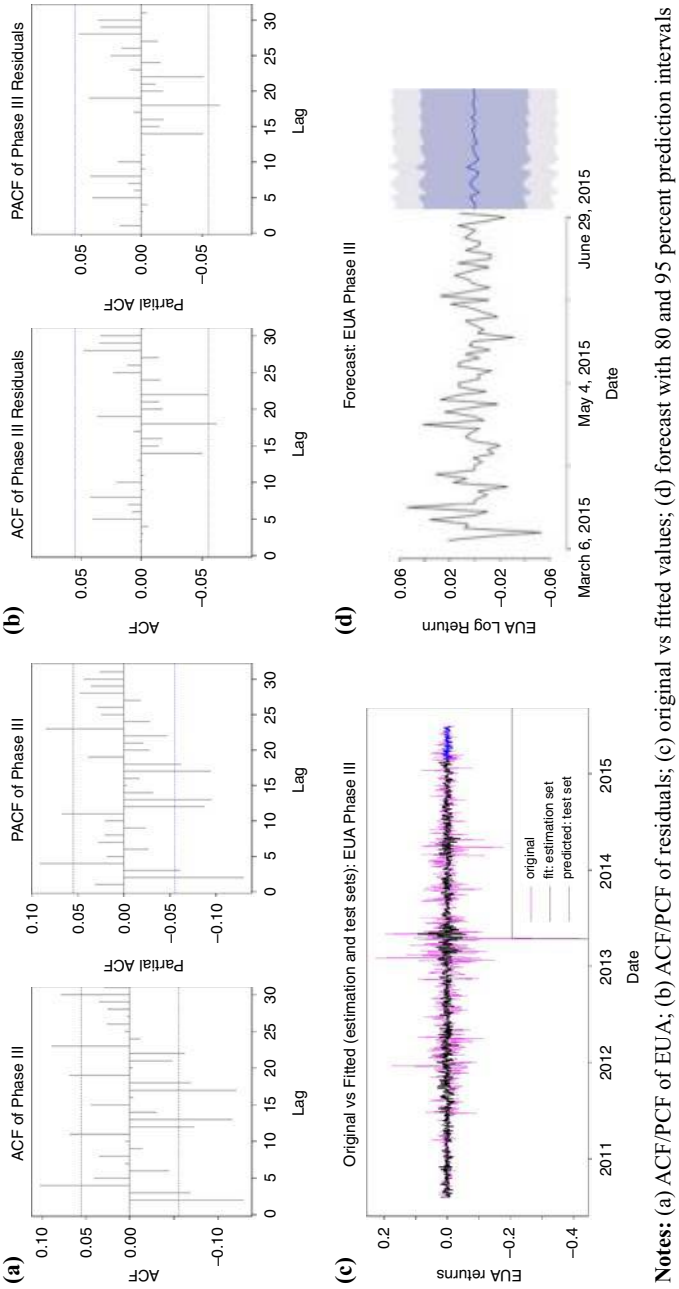

Notes: (a) ACF/PCF of EUA; (b) ACF/PCF of residuals; (c) original vs fitted values; (d) forecast with 80 and 95 percent prediction intervals

Figure 10. ARIMA errors model for EUA Phase III

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When X_t and N_t are assumed to follow the ARMA model, Equation (26) is known as the ARMAX model (Peter and Silvia, 2012). This ARMAX model is quite different from the ARMA model, because we work with two different series X_t and EUA_t – output series EUA_t is related to input series X_t. The "arimax" command in R works like the "arima" command, but allows inclusion of covariates. The covariates are centered first by subtracting the mean of the series, e.g. $brent_Z = brent - mean(brent)$. The argument "xtransf" is followed by a data frame in which each column corresponds to a covariate time series (same number of observations as the dependent variable time series). The argument transfer is followed by a list comprising one two-dimensional vector for each covariate specified by "xtransf." Similar to ARIMA modeling, iterative procedure is again adopted to find out the various models with significant lags of AE and lagged predictors as overall predictors. All possible models (based on ACF/PACF plots of EUA) are examined to verify the white noise character of residuals using ACF plots of residuals and the Portmanteau test of the residuals. Best model is then selected out of all verified (for the white noise character of residuals) models, which gives the least BIC. The best fit models in this study have the argument "list (c(0, 0))" implying that the covariate shall be included as it stands (no lagging, no filtering). Giving an argument $c(r, s)$ where both r and s are > 0 will enter the term $(\omega(1-\omega_1 B - \ldots - \omega_s B^s))/(1-\delta_1 B - \ldots - \delta_s B^s)$ into the model. Since covariates of best models for all the four series, namely EUA Phases I, II, II and continuous series have emerged with c(0, 0) specification so the term included will be ω covariate. The estimated model for all the four series can be written as follows:

$$EUA_t = Intercept_t + \omega\ Covariate + ARIMA\ Terms. \tag{27}$$

The model thus obtained is given in Table XVIII.

Table XVIII shows that all the AR coefficients, MA coefficients and the coefficients of covariates are significantly different from zero and neither the sum of AR coefficients nor the sum of MA coefficients is close to unity so there are no signs of unit roots. On the whole, this appears to be a good model. The orders of transfer functions for all the covariates are identified as (0,0), which means there are no AR or MA terms of the covariates in the models. We found MA1, MA11, MA13 and Brent as the significant predictors for EUA Phase I; AR1, AR6, AR7, AR18, RC1, Brent and DAX as significant predictors for EUA Phase II; AR2, AR4, AR13, AR17, MA11, MA12, MA23, RC2, Brent, coal and DAX as significant predictors

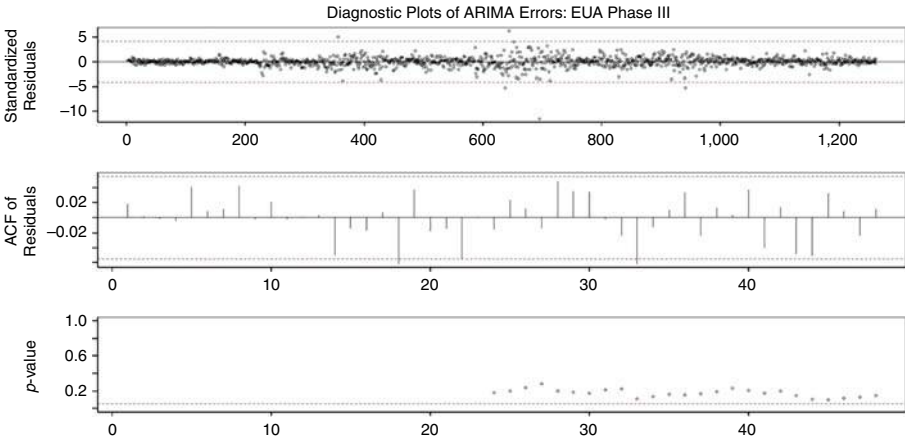

Figure 11. ARIMA errors model for EUA Phase III: diagnostic plots: standardized residuals, ACF of residuals, and the p-values of Portmanteau test

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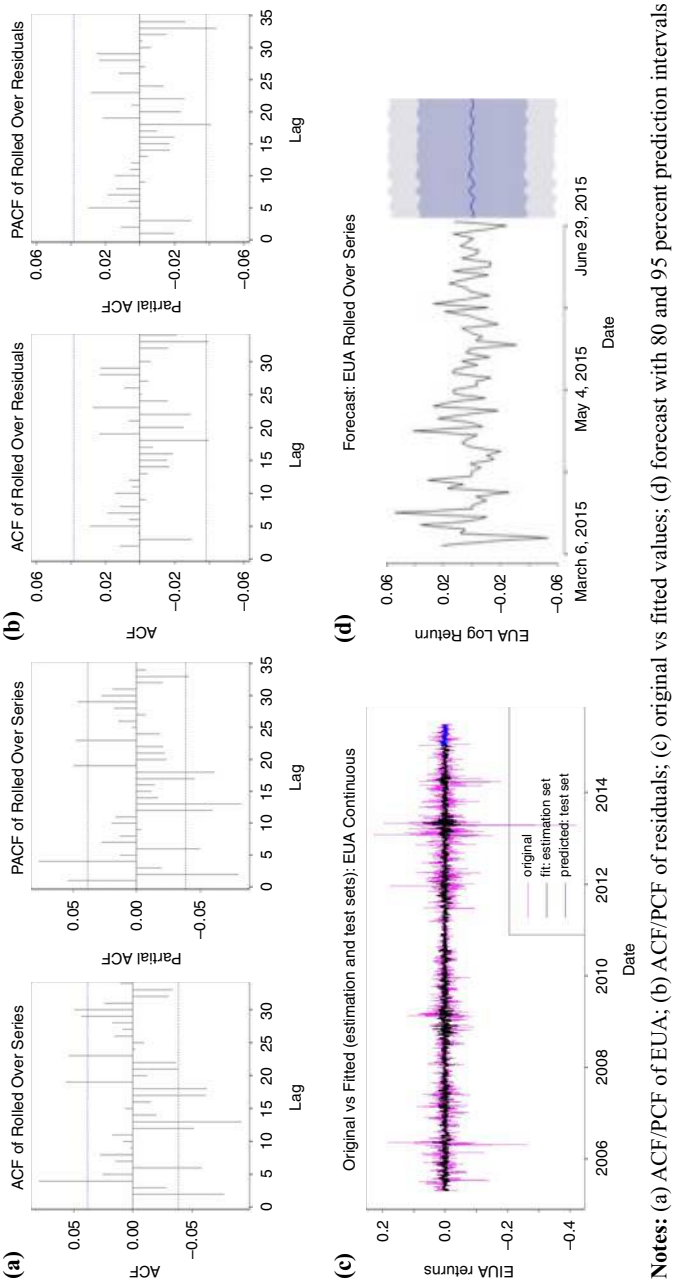

Notes: (a) ACF/PCF of EUA; (b) ACF/PCF of residuals; (c) original vs fitted values; (d) forecast with 80 and 95 percent prediction intervals

Figure 12. ARIMA errors model for EUA continuous series

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for EUA Phase III; and AR1, AR2, AR4, AR6, AR13, AR17, MA12, MA17, RC2, Brent, coal, DAX and temphot as significant predictors for EUA continuous series.

5.8 Overall comparison of models

We use the criteria of maximum adjusted R^2 to compare the linear models, minimum BIC value to compare various ARIMA models and that of minimum RMSE for overall comparison of all the models. Table XIX shows that the best linear model for predicting the returns of EUA Phase I is DLRLP with adjusted R^2 value of 0.2413 which means it could explain 24.13 percent of the total variation of EUA returns of Phase I. Though this value is too low to have precise prediction intervals, the significant predictors of the model can still be used to assess their effects on the EUA Phase I return series. Among ARIMA models, AE models with the BIC value of −1,172.81 is the best model. Overall, DLRLP model with estimation set RMSE value of 0.09567 and test set RMSE value of 0.09750 is the best model for EUA Phase I returns. The significant predictors found for EUA Phase I are RC1, RC2, napI, Brent, coal and DAX.

Table XX shows that the best linear model for predicting the returns of EUA Phase II is DLRLP with adjusted R^2 value of 0.1083 which means it could explain 10.63 percent of the total variation of EUA returns of Phase II. In this case, also the low value of R^2 will not entail precise prediction intervals, but the significant predictors can still be used to assess their effects on the EUA Phase II return series. Among the ARIMA models, AE models with the BIC value of −8,734.77 is the best model. Overall, the DLRLP model with the estimation set RMSE value of 0.02571 and test set RMSE value of 0.03997 is the best model for EUA Phase II returns. The significant predictors found for EUA Phase II are RC1, RC2, napI, napII, brent, coal, dax, temphot and tempcold.

Table XXI shows that the best linear model for predicting the returns of EUA Phase III is again DLRLP with adjusted R^2 value of 0.1241 which means the model could explain 12.41 percent of the total variation of EUA returns of Phase III. The interpretation of low R^2 is the same as in the results of earlier two phases, i.e. imprecise prediction intervals, but significant predictors. Among the ARIMA models, AE models with the BIC value of −4,998.85 is the best model. Overall, DLRLP model with an estimation set RMSE value of 0.03191 and test set RMSE value of 0.05735 is the best model for EUA Phase III returns. The significant predictors found for EUA Phase III are RC1, RC2, napIII, coal and dax.

Table XXII shows that the best linear model for predicting the returns of EUA rollover series is DLRLP with adjusted R^2 value of 0.0795 which means it could explain 7.95 percent of the total variation of EUA returns rollover series. Again this value is too low to have

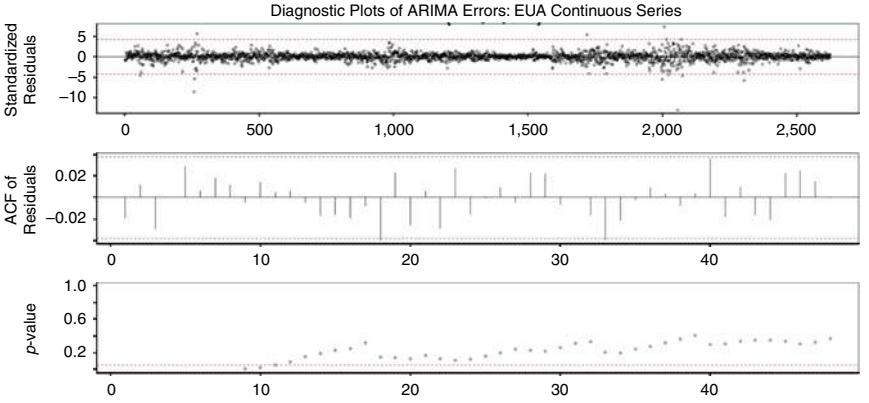

Figure 13. ARIMA errors model for EUA continuous series: diagnostic plots: standardized residuals, acf of residuals and the *p*-values of Portmanteau test

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	Estimate	SE	t value	Pr(> |t|)
Phase I (2005-2007)				
ma1	−0.1810	0.0365	−4.95	< 0.001***
ma11	0.3303	0.0356	9.26	< < 0.001***
ma13	−0.1932	0.0422	−4.57	< 0.001***
brent	0.7237	0.2102	3.44	< 0.001***
Model fitness criteria				
AIC_c	−1,195.34	BIC	−1,172.81	
RMSE	0.09971	MAE	0.04930	
Box-Ljung $\chi^2(32)$	33.14	p-value	0.411	
ΣAR	0	ΣMA	−0.0438	
Phase II (2008-2012)				
ar1	0.0792	0.0223	3.54	< 0.001***
ar6	−0.0825	0.0224	−3.68	< 0.001***
ar7	0.0624	0.0225	2.77	0.005**
ar18	−0.0653	0.0225	−2.89	0.003**
RC1	0.0037	0.0005	6.30	< 0.001***
brent	0.1651	0.0299	5.52	< 0.001***
dax	0.2783	0.0409	6.80	< 0.001***
Model fitness criteria				
AIC_c	−8,779.4	BIC	−8,734.77	
RMSE	0.02607	MAE	0.01802	
Box-Ljung $\chi^2(29)$	40.76	p-value	0.072	
ΣAR	−0.0061	ΣMA	0	
Phase III (2013-2015)				
ar2	−0.1115	0.0277	−4.01	< 0.001***
ar4	0.0799	0.0279	2.86	0.004**
ar13	−0.1018	0.0275	−3.69	< 0.001***
ar17	−0.1047	0.0278	−3.75	< 0.001***
ma3	−0.0856	0.0280	−3.05	0.002**
ma11	0.0678	0.0294	2.30	0.021*
ma12	−0.0881	0.0280	−3.14	0.001**
ma23	0.1019	0.0270	3.76	< 0.001***
RC2	0.0042	0.0008	4.81	< 0.001***
brent	0.1308	0.0569	2.29	0.021*
coal	−0.2259	0.0849	−2.66	0.007**
dax	0.2591	0.0731	3.54	< 0.001***
Model fitness criteria				
AICc	−5,065.39	BIC	−4,998.85	
RMSE	0.03217	MAE	0.02096	
Box-Ljung $\chi^2(24)$	35.53	p-value	0.060	
ΣAR	−0.2381	ΣMA	−0.0039	
Continuous (2005-2015)				
ar1	0.0733	0.0166	4.40	< 0.001***
ar2	−0.0669	0.0170	−3.92	< 0.001***
ar4	0.0644	0.0174	3.69	< 0.001***
ar6	−0.0498	0.0170	−2.92	0.003**
ar13	−0.0834	0.0176	−4.73	< 0.001 ***
ar17	−0.4792	0.0667	−7.18	< 0.001***
ma12	−0.05	0.0173	−3.2053	0.001**
ma17	0.4505	0.0693	6.50	< 0.001***

(*continued*)

Table XVI. ARIMA errors model for EUA

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	Estimate	SE	t value	Pr(> |t|)
RC2	0.0033	0.0005	5.77	< 0.001***
brent	0.1555	0.0301	5.15	< 0.001***
coal	−0.0941	0.0401	−2.34	0.0191*
dax	0.2523	0.0424	5.94	< 0.001***
temphot	−0.0081	0.0039	−2.07	0.037*
Model fitness criteria				
AIC_c	−11,057.81	BIC	−10,975.76	
RMSE	0.02926	MAE	0.01936	
Box-Ljung $\chi^2(23)$	31.98	p-value	0.10	
ΣAR	−0.5417	ΣMA	0.3949	

Table XVI. **Note:** *,**,***Significant at 0.05, 0.01 and 0.001 levels, respectively

precise prediction intervals. Among the ARIMA models, AE model with a BIC value of −10975.76 is the best model. Overall, AE model with an estimation set RMSE value of 0.02926 and test set RMSE value of 0.06169 is the best model for EUA rollover series returns. The significant predictors found for EUA continuous series are RC2, brent, coal, dax and temphot. Therefore, RC1, RC2, napI, napII, napIII, brent, coal, dax, temphot and tempcold have been verified as price determinants in one phase or the other for EUA returns. Since price relationship is intricately related with the volatility relationship, further study needs to be done by taking volatility into consideration and by taking different kinds of models like artificial neural networks and GARCH models to arrive at the final predictive models.

6. Discussion

The EU ETS is proceeding in three phases. The pilot phase (Phase I) went from 2005 to 2007, and is usually termed as a "warm-up" period prior to the full-fledged introduction of the Kyoto Protocol. Phase II went from 2008 to 2012, alongside with the Kyoto Protocol (the UN policy framework till date), and the third Phase from 2013 onwards aims to reduce emissions by 20 percent by 2020. This study uses ECX December 2007, December 2012 and December 2015 expiry contracts for phase analysis and the continuous series December 2015 contract for continuous analysis. The predictor variables considered are NAP announcements, energy variables (Brent oil, coal and natural gas), averaged extreme Europe temperature and German Stock Exchange index (DAX) as a daily indicator of economic activity. There are 682, 1,975, 1,263 Announcements, Energy variables (Brent Oil, Coal, and Natural gas), averaged extreme and 2,625 observations for Phase I, Phase II, Phase III and continuous series, respectively. Since, the predictor variables and the predicted variable data series are found to be non stationary, all these series are first differenced before analysis. In order to get rid of multicollinearities, PCA is adopted to arrive at the independent factors and predictor variables. The iterative procedure is adopted to find the best model for all four EUA returns series in each category of the seven modeling techniques by using suitable metrics like adjusted R^2 and BIC. Overall best model was arrived at for each EUA return series by using the metric of RMSE for the test set.

The DLRLP model with estimation set RMSE value of 0.09567 and test set RMSE value of 0.09750 is the best model for EUA Phase I returns. The significant predictors found for EUA Phase I are EUA Phase I (Lags 1, 10, 11, 12, 13, 17 and 18), RC1 (lags 11 and 12), RC2 (lags 12 and 20), napI (lags 6 and 10), Brent, coal (lags 18 and 19) and dax (lag 8). The DLRLP model with an estimation set RMSE value of 0.02571 and test set RMSE value of 0.03997 is the best model for EUA Phase II returns. The significant predictors found for

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	Estimate	SE	t value	Pr(> |t|)
Phase I (2005-2007)				
ma1	−0.1810	0.0365	−4.95	< 0.001***
ma11	0.3303	0.0356	9.26	< 0.001***
ma13	−0.1932	0.0422	−4.57	< 0.001***
brent	0.7237	0.2102	3.44	< 0.001***
Model fitness criteria				
AIC_c	−1,190.17	BIC	−1,167.66	
RMSE	0.09983	MAE	0.04934	
Box-Ljung $\chi^2(32)$	33.08	p-value	0.414	
ΣAR	0	ΣMA	−0.0437	
Phase II (2008-2012)				
ar1	0.0792	0.0223	3.53	< 0.001***
ar6	−0.0821	0.0224	−3.66	< 0.001***
ar7	0.0616	0.0225	2.73	0.006**
ar18	−0.0665	0.0225	−2.94	0.003**
RC1	0.0037	0.0005	6.27	< 0.001***
brent	0.1670	0.0299	5.58	< 0.001***
dax	0.2771	0.0409	6.77	< 0.001***
Model fitness criteria				
AIC_c	−8,771.01	BIC	−8,726.39	
RMSE	0.02607	MAE	0.01801	
Box-Ljung $\chi^2(29)$	40.96	p-value	0.069	
ΣAR	−0.0077	ΣMA	0	
Phase III (2013-2015)				
ar2	−0.1115	0.0277	−4.01	< 0.001***
ar4	0.0799	0.0279	2.85	0.004**
ar13	−0.1017	0.0276	−3.68	< 0.001***
ar17	−0.1047	0.0279	−3.75	< 0.001***
ma3	−0.0856	0.0281	−3.04	0.002**
ma11	0.0677	0.0294	2.29	0.021*
ma12	−0.0880	0.0280	−3.14	0.001**
ma23	0.1020	0.0271	3.76	< 0.001***
RC2	0.0042	0.0008	4.78	< 0.001***
brent	0.1307	0.0570	2.29	0.021*
coal	−0.2261	0.0850	−2.66	0.007**
dax	0.2591	0.0732	3.53	< 0.001***
Model fitness criteria				
AIC_c	−5,055.34	BIC	−4,988.83	
RMSE	0.03219	MAE	0.02099	
Box-Ljung $\chi^2(24)$	49.30	p-value	0.068	
ΣAR	−0.2381	ΣMA	−0.0040	
Continuous (2005-2015)				
ar1	0.0736	0.0166	4.41	< 0.001***
ar2	−0.0666	0.0170	−3.91	< 0.001***
ar4	0.06391	0.0174	3.66	< 0.001***
ar6	−0.0497	0.0170	−2.92	0.003**
ar13	−0.0834	0.0176	−4.73	< 0.001***
ar17	−0.4805	0.0666	−7.20	< 0.001***
ma12	−0.055	0.0173	−3.19	0.001**
ma17	0.4517	0.0692	6.52	< 0.001***

(*continued*)

Table XVII. ARIMA errors with lagged predictors model for EUA

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	Estimate	SE	t value	Pr(> |t|)
RC2	0.0033	0.0005	5.76	< 0.001***
brent	0.1569	0.0301	5.20	< 0.001***
coal	−0.0950	0.0401	−2.36	0.018*
dax	0.2518	0.0424	5.92	< 0.001***
temphot	−0.0081	0.0039	−2.07	0.038*
Model fitness criteria				
AIC_c	−11,049.09	BIC	−10,967.05	
RMSE	0.02926	MAE	0.01936	
Box-Ljung $\chi^2(23)$	32.08	p-value	0.098	
ΣAR	−0.5428	ΣMA	0.3964	

Note: *,**,***Significant at 0.05, 0.01 and 0.001 levels, respectively

Table XVII.

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EUA Phase II are EUA Phase II (lags 1, 6, 7, 11 and 18), RC1, RC1 (lag 3), napI (lags 1 and 11), napII (lags 9 and 14), brent, coal (lag 1), dax, temphot (lag 18) and tempcold (lag 15). The DLRLP model with an estimation set RMSE value of 0.03191 and test set RMSE value of 0.05735 is the best model for EUA Phase III returns. The significant predictors found for EUA Phase III are EUA Phase III (lags 2, 3, 4, 12, 13, 17 and 18), RC1 (lags 4 and 15), RC2, RC2 (lag 19), napIII (lags 11 and 20), coal (lag 11), dax and dax (lags 6 and 17). The AE model with estimation set RMSE value of 0.02926 and test set RMSE value of 0.06169 is the best model for EUA continuous series. The significant predictors found for EUA continuous series are AR1, AR2. AR4, AR6, AR13, AR17 (All auto regressive terms), MA12, MA17 (both moving average terms), RC2, brent, coal, dax and temphot. Therefore, RC1, RC2, napI, napII, napIII, brent, coal, dax, temphot and tempcold are identified as price determinants for EUA returns.

Though the adjusted R^2 value for all the EUA return series is very low, yet the interpretations of the significant variables are the same as they are for high adjusted R^2 models. Even when adjusted R^2 is low, low p-values still indicate a real relationship between the significant predictors and EUA returns. Regardless of the adjusted R^2, the significant coefficients still represent a mean change in the EUA returns for one unit of change in the predictor while holding other predictors in the model constant. Obviously, this type of information is extremely valuable, especially in this new market. Adjusted R^2 neither indicates the adequacy of a regression model nor determines whether the coefficient estimates and predictions are biased, which is why we assess the residual plots in this study. The residual plots of all the models give white noise character which ensures that the model is adequate and well specified. A low adjusted R^2 is most problematic when we want to produce predictions that are reasonably precise (have a small enough prediction interval), which is not the case here. The additional predictors may not enhance the true explanatory power of the model, if the data contain an inherently higher amount of unexplainable variability.

Konar and Cohen (2001) relate the valuation of the firms in the S&P500 to objective measures of their environmental performance. After controlling for variables historically believed to explain financial performance, they find that poor environmental performance is negatively correlated with the intangible asset value of the firms. In modern times, the companies need to consider environmental issues such as trading in carbon allowances (or permits), investment in low CO_2 emission technologies, counting the costs of carbon regularity compliance and passing on the increased cost of carbon regulation to consumers through higher prices for proper profit analysis and valuation. Ratnatunga and Balachandran (2013) talk about the impact of the Kyoto Protocol mechanisms on the changing information paradigms of cost and managerial accounting. Under various carbon

	Estimate	se	t value	Pr(> |t|)
Phase I (2005-2007)				
ma1	−0.1824	0.0365	−4.98	< 0.001***
ma11	0.3292	0.0357	9.22	< < 0.001***
ma13	−0.1948	0.0423	−4.60	< 0.001***
brent-MA0	0.7468	0.2103	3.55	< 0.001***
Model fitness criteria				
RMSE	0.09966	MAE	0.0493	
Box-Ljung χ^2(32)	33.11	p-value	0.41	
ΣAR	0	ΣMA	−0.0480	
Phase II (2008-2012)				
ar1	0.0789	0.0223	3.52	< 0.001***
ar6	−0.0826	0.0224	−3.68	< 0.001***
ar7	0.0620	0.0225	2.75	0.005**
ar18	−0.0656	0.0225	−2.90	0.003**
RC1−MA0	0.0037	0.0005	6.29	< 0.001***
brent-MA0	0.1665	0.0299	5.56	< 0.001***
dax-MA0	0.2788	0.0409	6.81	< 0.001***
Model fitness criteria				
RMSE	0.02607	MAE	0.01802	
Box-Ljung χ^2(29)	40.81	p-value	0.071	
ΣAR	−0.0073	ΣMA	0	
Phase III (2013-2015)				
ar2	−0.1119	0.0277	−4.03	< 0.001***
ar4	0.0796	0.0279	2.84	0.004**
ar13	−0.1021	0.0275	−3.70	< 0.001***
ar17	−0.1050	0.0278	−3.76	< 0.001***
ma3	−0.0861	0.0280	−3.06	0.002**
ma11	0.0673	0.0294	2.28	0.022*
ma12	−0.0885	0.0280	−3.16	0.001**
ma23	0.1017	0.0270	3.75	< 0.0010***
RC-MA0	0.0042	0.0008	4.81	< 0.001***
brent-MA0	0.1302	0.0569	2.28	0.022*
coal-MA0	−0.2279	0.0849	−2.68	0.007**
dax-MA0	0.2624	0.0731	3.58	< 0.001***
Model fitness criteria				
RMSE	0.0321	MAE	0.02096	
Box-Ljung χ^2(24)	49.45	p-value	0.066	
ΣAR	−0.2394	ΣMA	−0.0056	
Continuous (2005-2015)				
ar1	0.0734	0.0166	4.40	< 0.001***
ar2	−0.0669	0.0170	−3.92	< 0.001***
ar4	0.0643	0.0174	3.69	< 0.001***
ar6	−0.0497	0.0170	−2.91	0.003**
ar13	−0.0835	0.0176	−4.73	< 0.001***
ar17	−0.4793	0.0667	−7.18	< 0.001***
ma12	−0.0555	0.0173	−3.20	0.001***
ma17	0.4507	0.0693	6.50	< 0.001***
RC2-MA0	0.0033	0.0005	5.76	< 0.001***

(*continued*)

Table XVIII. ARIMAX model for EUA

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	Estimate	se	t value	Pr(> |t|)
brent-MA0	0.1548	0.0301	5.13	< 0.001***
coal-MA0	−0.0940	0.0401	−2.34	0.019*
dax-MA0	0.2546	0.0424	5.99	< 0.001***
temphot-MA0	−0.0076	0.0039	−1.93	0.053
Model fitness criteria				
RMSE	0.0292	MAE	0.0193	
Box-Ljung $\chi^2(23)$	31.99	p-value	0.100	
ΣAR	−0.5417	ΣMA	0.3951	

Note: *,**,***Significant at 0.05, 0.01 and 0.001 levels, respectively

Table XVIII.

Table XIX. Comparative summary of models of EUA phase I

	Linear regression	Dynamic linear regression	Dynamic linear regression with lags of predictors	ARIMA	ARIMA errors	ARIMA errors with lagged predictors	ARIMAX
R^2	0.0136	0.1487	*0.2609*	–	–	–	–
Adjusted R^2	0.0121	0.1369	*0.2413*	–	–	–	–
AIC_c	–	–	–	−1,185.66	*−1,195.34*	−1,190.17	–
BIC	–	–	–	−1,167.62	*−1,172.81*	−1,167.66	–
RMSE (estimation set)	0.09939	0.09936	*0.09567*	0.10058	0.09971	0.09983	0.09966
MAE (estimation set)	*0.04161*	0.05136	0.05296	0.04847	0.04930	0.04934	0.04934
RMSE (test set)	0.10561	0.13865	*0.09750*	0.14306	0.10587	0.10669	0.10525
MAE (test set)	*0.04549*	0.08012	0.09282	0.07159	0.05631	0.05562	0.05849

Table XX. Comparative summary of models of EUA Phase II

	Linear regression	Dynamic linear regression	Dynamic linear regression with lags of predictors	ARIMA	ARIMA errors	ARIMA errors with lagged predictors	ARIMAX
R^2	0.0712	0.0930	*0.1156*	–	–	–	–
Adjusted R^2	0.0693	0.0889	*0.1083*	–	–	–	–
AIC_c	–	–	–	−8,635.45	*−8,779.4*	−8,771.01	–
BIC	–	–	–	−8,613.12	*−8,734.77*	−8,726.39	–
RMSE (estimation set)	0.02630	0.02603	*0.02571*	0.02709	0.02607	0.02607	0.02607
MAE (estimation set)	*0.01797*	0.01799	0.01807	0.01891	0.01802	0.01801	0.01802
RMSE (test set)	0.04627	0.04430	*0.03997*	0.05131	0.04865	0.04868	0.04868
MAE (test set)	*0.01943*	0.01957	0.01985	0.02062	0.01979	0.01977	0.01977

emissions trading schemes all over the world (including regional ones), organizations will need to implement carbon management schemes to meet carbon emission targets, earn revenue and reduce costs all simultaneously. So, carbon management has emerged as a newer and vastly impacting determinant of financial performance and strategic direction of the firms. In this regard, the current study contributes to the effective carbon management by finding out the best method to forecast the carbon credits and by isolating the price determinants of carbon credits.

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	Linear regression	Dynamic linear regression	Dynamic linear regression with lags of predictors	ARIMA	ARIMA errors	ARIMA errors with lagged predictors	ARIMAX
R^2	0.0353	0.0494	*0.1361*	–	–	–	–
Adjusted R^2	0.0322	0.0463	*0.1241*	–	–	–	–
AIC_c	–	–	–	−5,015.31	*−5,065.39*	−5,055.34	–
BIC	–	–	–	−4,969.19	*−4,998.85*	−4,988.83	–
RMSE (estimation set)	0.03349	0.03344	*0.03191*	0.03292	0.03217	0.03219	0.03216
MAE (estimation set)	0.02115	0.02139	0.02129	0.02152	*0.02096*	0.02099	0.02096
RMSE (test set)	0.06196	0.06093	*0.05735*	0.05851	0.05827	0.05833	0.05821
MAE (test set)	0.04549	0.04416	0.04368	0.04405	*0.04135*	0.04169	0.04166

Table XXI. Comparative summary of models of EUA Phase III

	Linear regression	Dynamic linear regression	Dynamic linear regression with lags of predictors	ARIMA	ARIMA errors	ARIMA errors with lagged predictors	ARIMAX
R^2	0.0443	0.0825	*0.0844*	–	–	–	–
Adjusted R^2	0.0429	0.0779	*0.0795*	–	–	–	–
AIC_c	–	–	–	−10,935.13	*−11,057.81*	−11,049.09	–
BIC	–	–	–	−10,882.34	*−10,975.76*	−10,967.05	–
RMSE (estimation set)	0.02992	0.02936	0.02933	0.03001	*0.02926*	0.02926	0.02926
MAE (Estimation Set)	0.01942	*0.01935*	0.01950	0.02005	0.01936	0.01936	0.01936
RMSE (test set)	0.06328	0.06299	0.06293	0.06348	*0.06169*	0.06172	0.06189
MAE (test set)	0.04381	*0.04359*	0.04427	0.04593	0.04388	0.04391	0.04394

Table XXII. Comparative summary of models of EUA continuous series

Abatement policies should seek emission compliance at the lowest possible cost while the production portfolio should constitute more and more cleaner technologies. While the policies did realize the emission reductions but shortcomings like windfall gains and surplus supply, etc. created impediments for the appropriate price discovery of carbon permits. The solution being followed currently is 100 percent auctioning of allowances. Another solution could be allocating the allowances proportionally to the instantaneous production of goods. The theoretical and quantitative analyses of cap-and-trade schemes for the purpose of emissions reduction are still in their infancy. We provide policy makers with quantitative tools to analyze price determinants and price discovery of carbon permits under various cap-and-trade schemes (generic cap-and-trade scheme with free allocations, cap-and-trade schemes with a 100 percent auctioning of allowances and schemes with allocating the allowances proportionally to the instantaneous production of goods) capable of achieving pollution targets at low reduction costs.

This study considers EUA data from all three phases and the continuous series to fit seven different predictive models for comprehensive analysis and adopt PCA to eliminate multicollinearities. The study establishes the DLRLP model as the best model for forecasting EUA Phase I, Phase II and Phase III returns and AE model for that of EUA continuous series returns. The study contributes to effective carbon management by providing the quantitative framework for analyzing cap-and-trade schemes and by finding out the best method for forecasting the carbon permits and isolating the price determinants of carbon permits. The present study is based on daily closing prices of EUA futures and

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intra-day variability is not taken into consideration. The study focuses on one instrument, i.e. EUA, and does not conduct the econometric analysis of other financial instruments like ERU. The present study paves the way for future research considering high-frequency data in order to have an estimate of intra-day volatility. Further, the secondary data could also be analyzed using other AI models like support vector machines, ensemble methods and fuzzy systems for estimation to get better insights. The present study could be extended to analyze EUA options, which might add to the knowledge of market mechanisms of EUA.

7. Conclusion

The best model for forecasting the EUA return series of all three phases is DLRLP and the best model for forecasting EUA continuous series is AE. The latent factors RC1, RC2 (capturing the effects of the variables CDS, CSS, SP and natural gas price) and the variables napI, napII, napIII (NAP announcements events), brent, coal (both energy variables), dax (German Stock Exchange index as an indicator of economic activity), temphot and tempcold (extreme temperature events) have been verified as the price determinants in one phase or the other for EUA returns. The study contributes to effective carbon management by providing the quantitative framework for analyzing cap-and-trade schemes.

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P.K. Jain is a Professor of Finance at the Department of Management Studies, IIT Delhi. He holds an MCom Degree as well as PhD Degree in Finance from the Delhi School of Economics, University of Delhi. He earlier served as the Head of Department of Management Studies and the Co-ordinator of Dalmia Research Programme on Management in Asia. He was the Modi Foundation Chair Professor as well as the Dalmia Chair Professor. He has a teaching experience of more than 40 years in subjects related to Management Accounting, Financial Management, Financial Analysis, Cost Analysis and Cost Control. His research is primarily focused in the area of corporate finance. He has authored/co-authored more than nine text books and 14 research books/monographs. AS an Empiricist, he investigates applied problems using real-business data to have the managerial implications of various financial aspects. P.K. Jain has published more than 160 research papers in journals of national and international repute. Recently, his paper on "Capital budgeting decisions: evidence from India' has been adjudged with the "Literati Award" by Emerald for Outstanding Excellence in Research. He has also presented research papers in more than three dozen conferences held in India and abroad; and many of them have been published in the conference proceedings. He is on the Board of academic institutions and reviewer of journals like, *Vision*, *Abhigyan*, *Vikalp*, *Finance India*, *Management and Change*. He has been invited by several institutions as a Visiting Professor to deliver lectures. The select list includes IIM Lucknow, IIM Indore, IIFT, ICAI, AIMA, IIITM, etc. He had been a Visiting Faculty at the International Centre for Promotion of Enterprises Ljubljana, Slovenia for more than a decade. He has been conferred Best Teacher Award by IIT Delhi in January 2012. In July 2012, he has also been conferred 3rd Asia's Best B-School Awards entitled "Best Professor in Financial Management" (in Singapore) by World Education Congress and CMO Asia as well as "Best Teacher in Financial Management" (at Mumbai) by *Amar Ujjala* (in November, 2012). He has organized several management developments Programme (MDPs)/Continuing Education Programmes (CEPs) both for public and private sector executives. He has been organizing (wef April 2001) Online Executive Development Programmes on Finance for Non-Finance executives. He has been associated with several research projects and consultancy in the field of finance.

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